On the precision of dynamic gene regulation

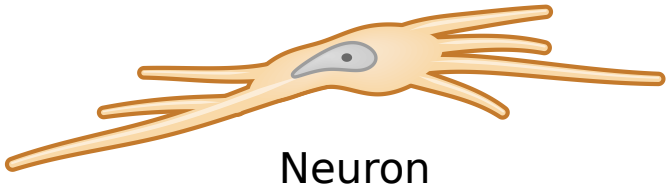
Tarek Tohme

Under the supervision of

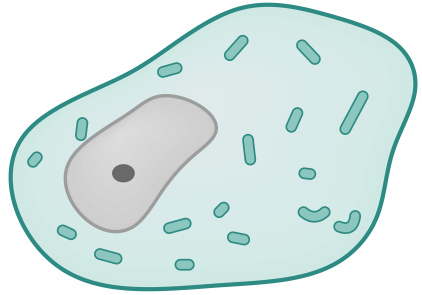
Aleksandra Walczak and Thierry Mora

November 4, 2026



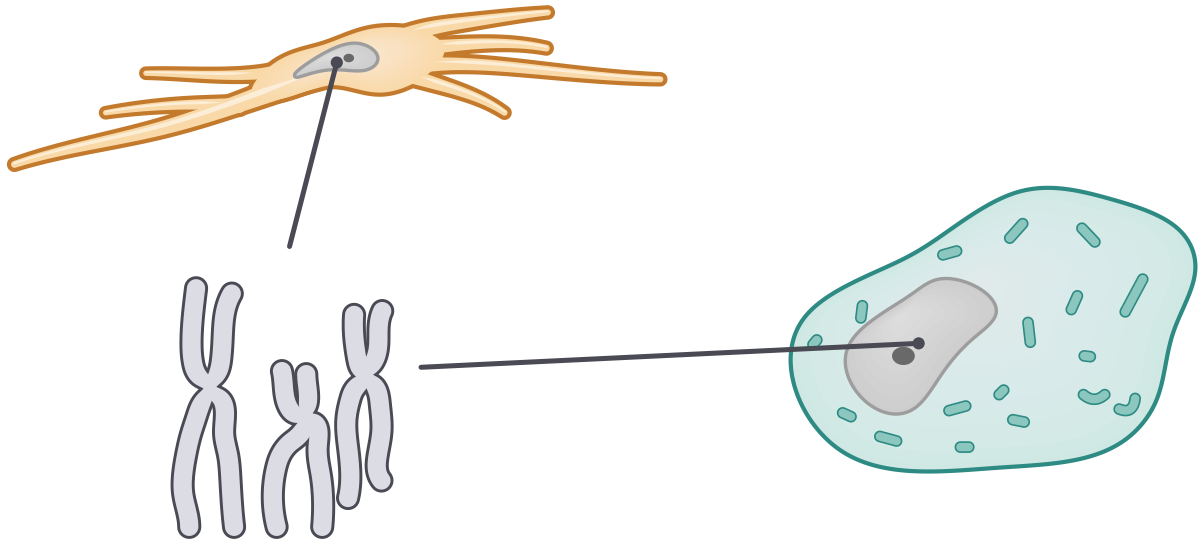


Neuron



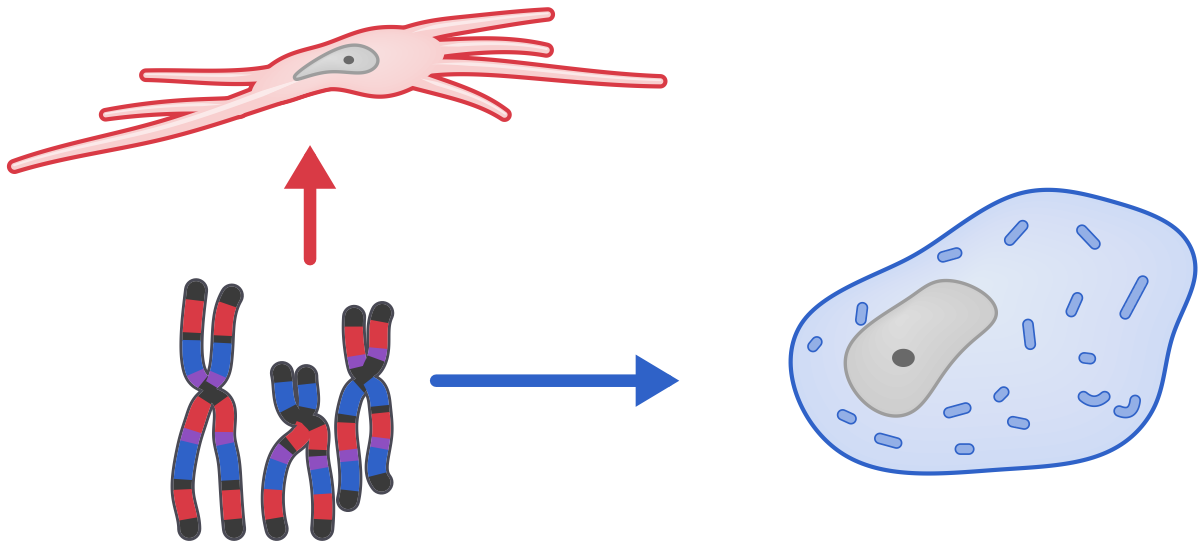
Kidney cell

scene1



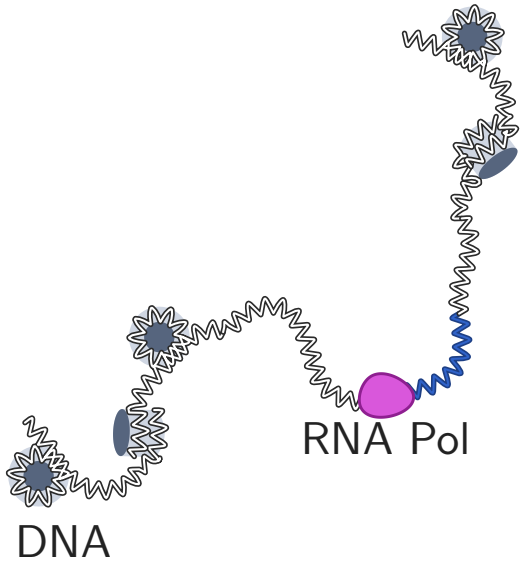
scene2

Gene expression is regulated



scene3

Overview of gene regulation

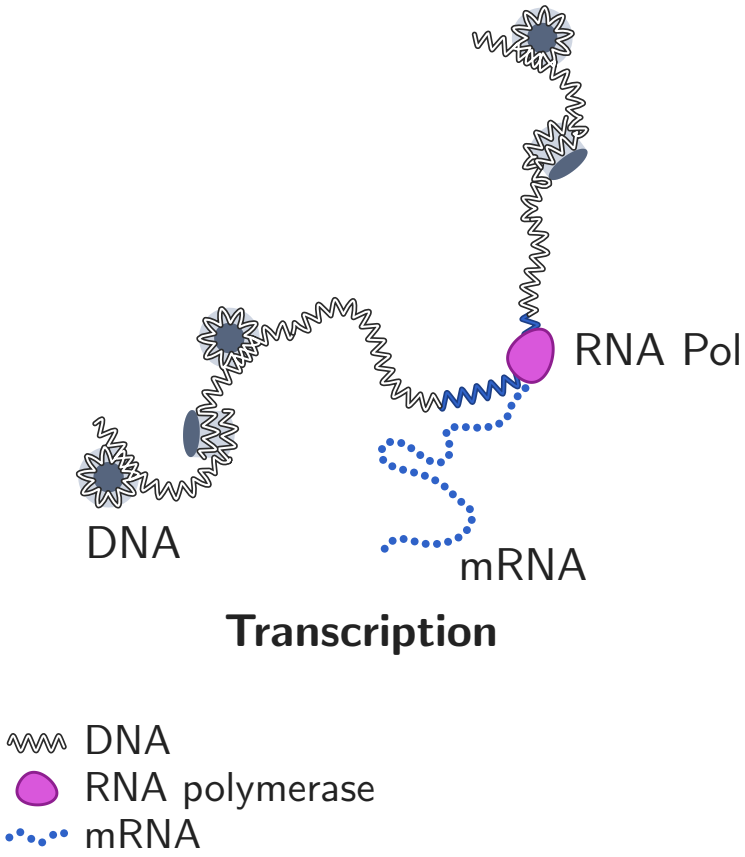


Transcription

DNA
RNA polymerase

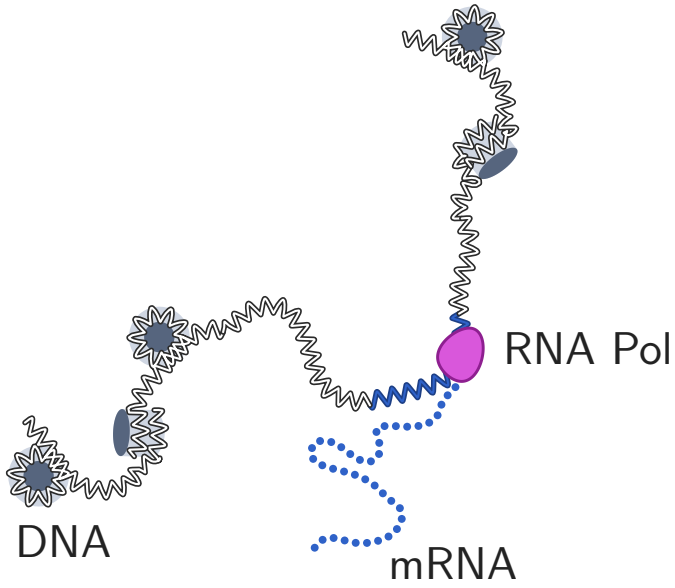
scene4_1

Overview of gene regulation

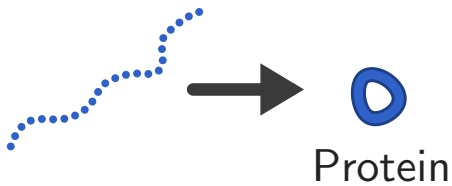


scene4_2

Overview of gene regulation



Transcription

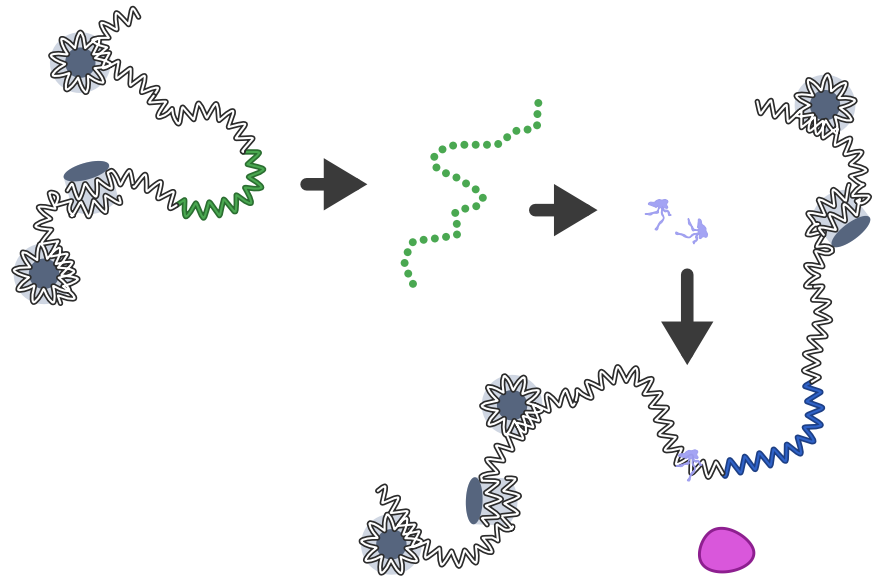


Translation

DNA
RNA polymerase
mRNA

scene4$_3$

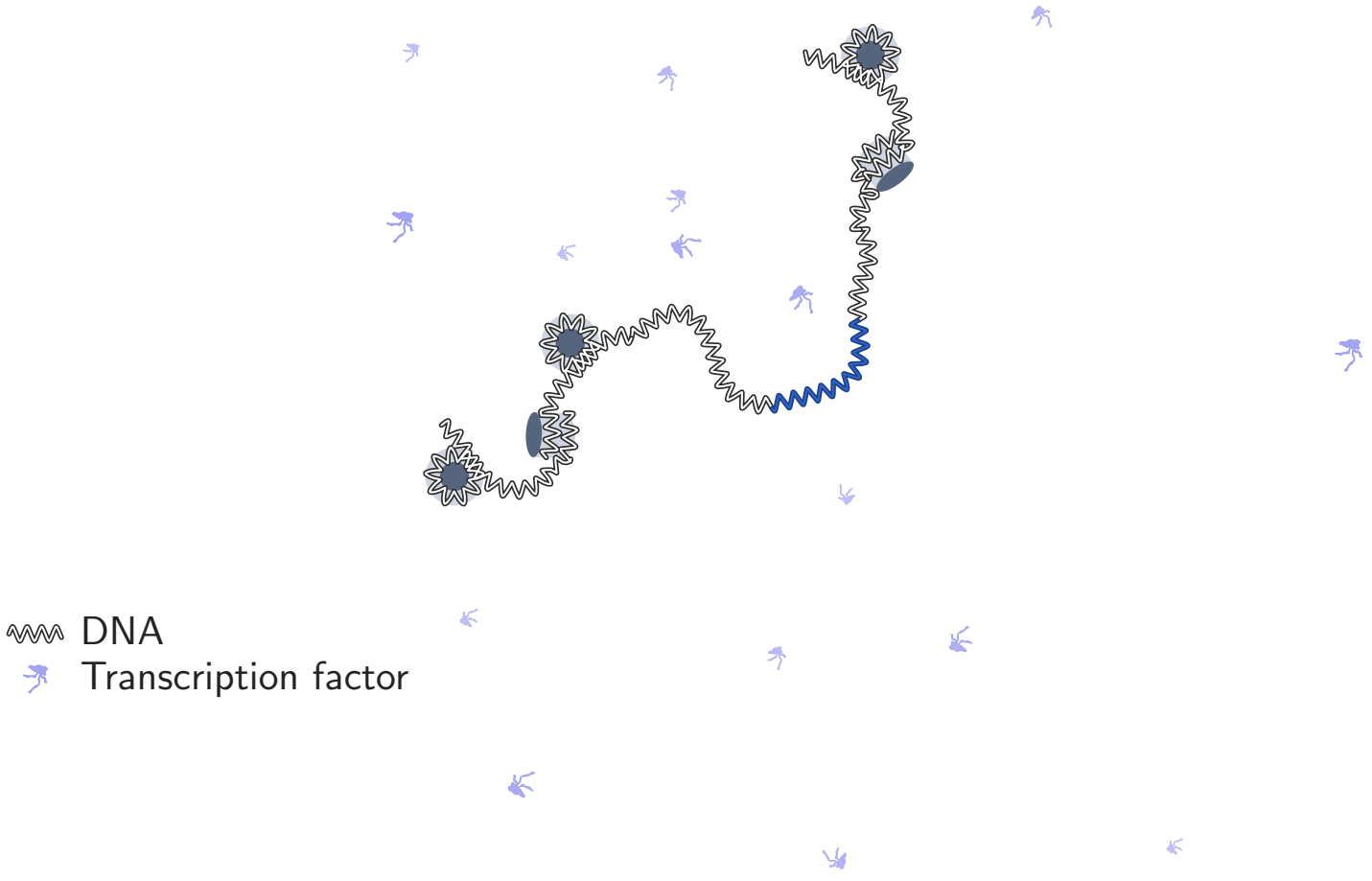
Overview of gene regulation



DNA
Transcription factor
RNA polymerase
mRNA

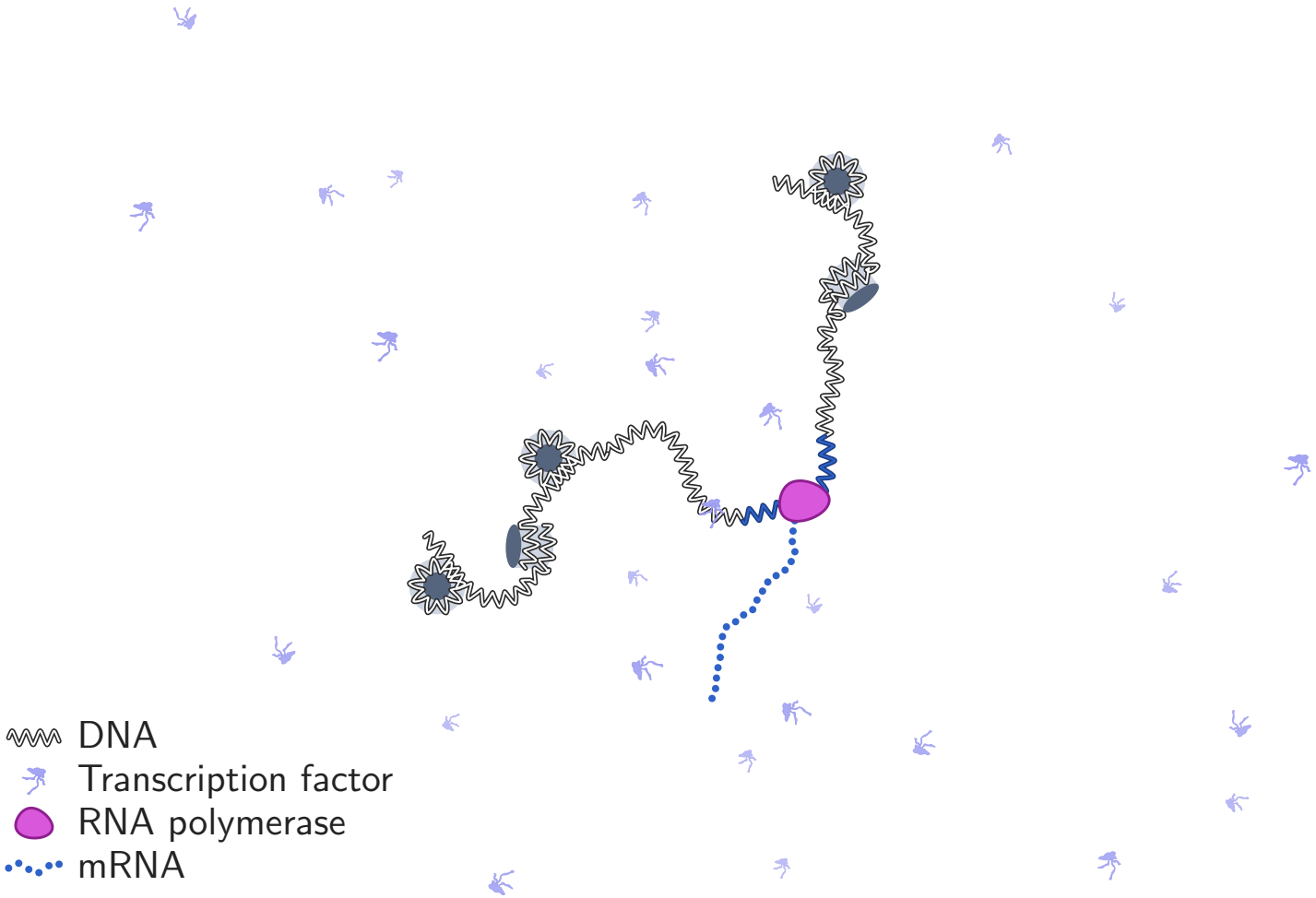
scene5

Expression levels are continuous ...



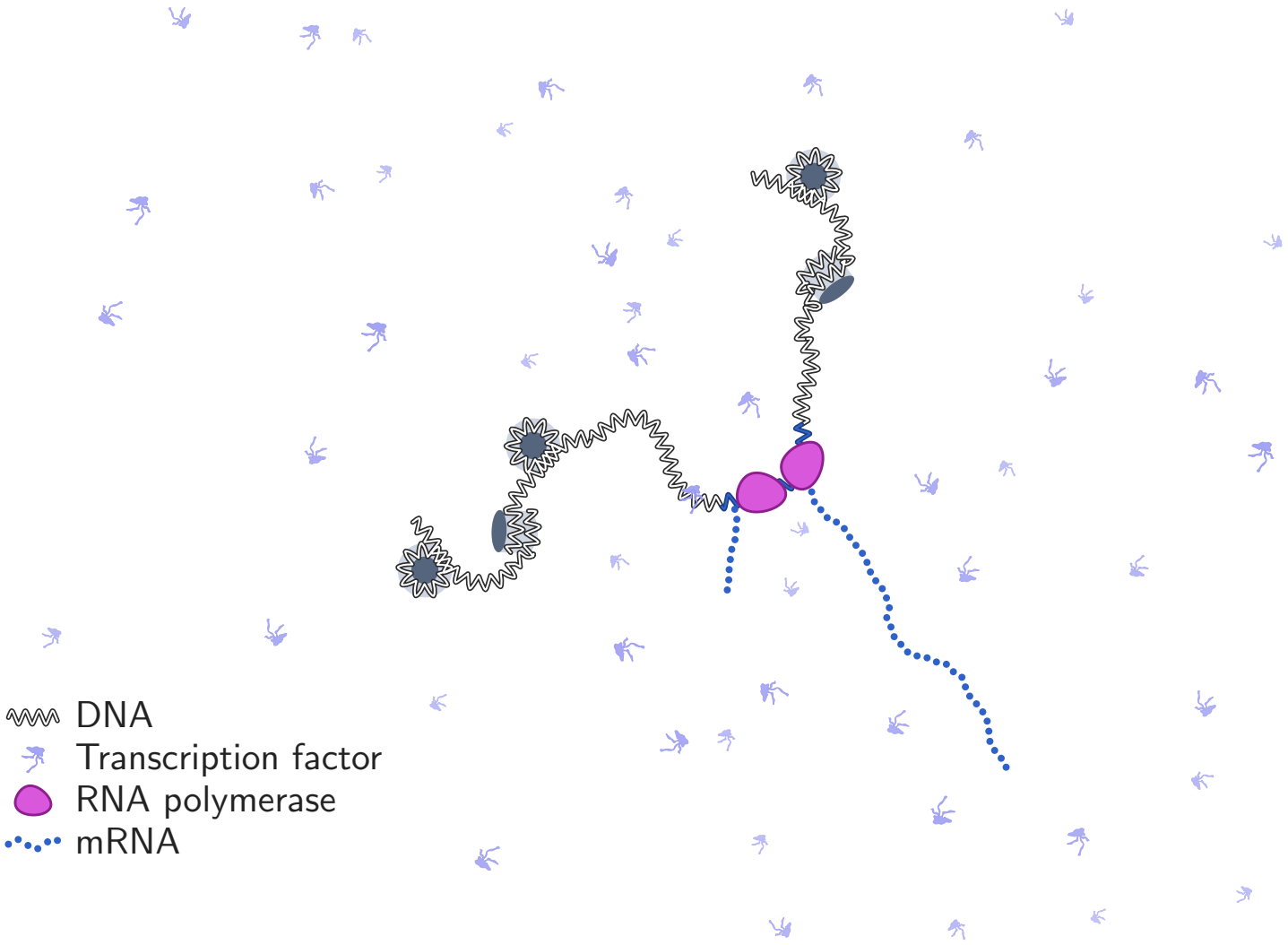
scene6$_1$

Expression levels are continuous ...



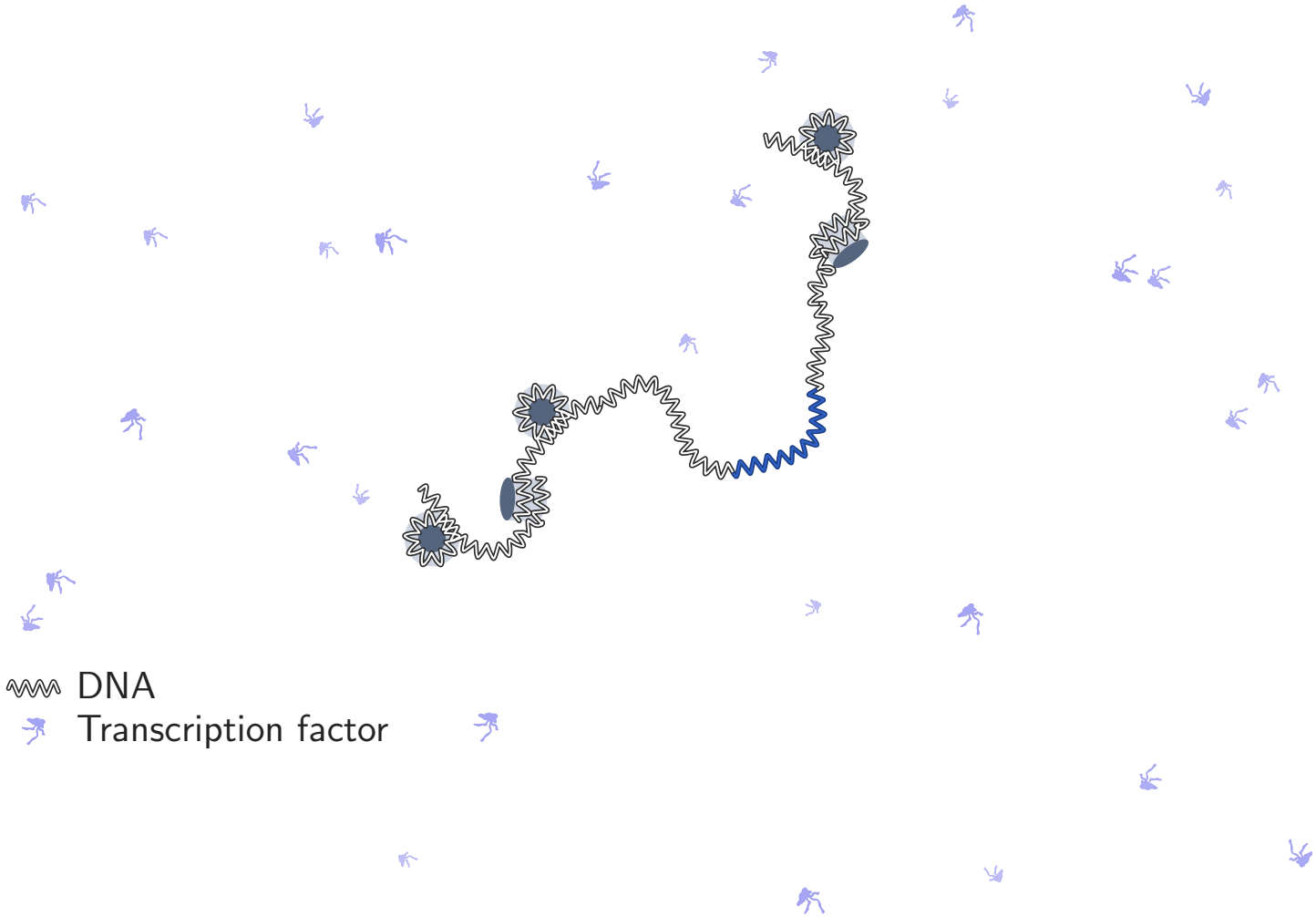
scene6_2

Expression levels are continuous ...



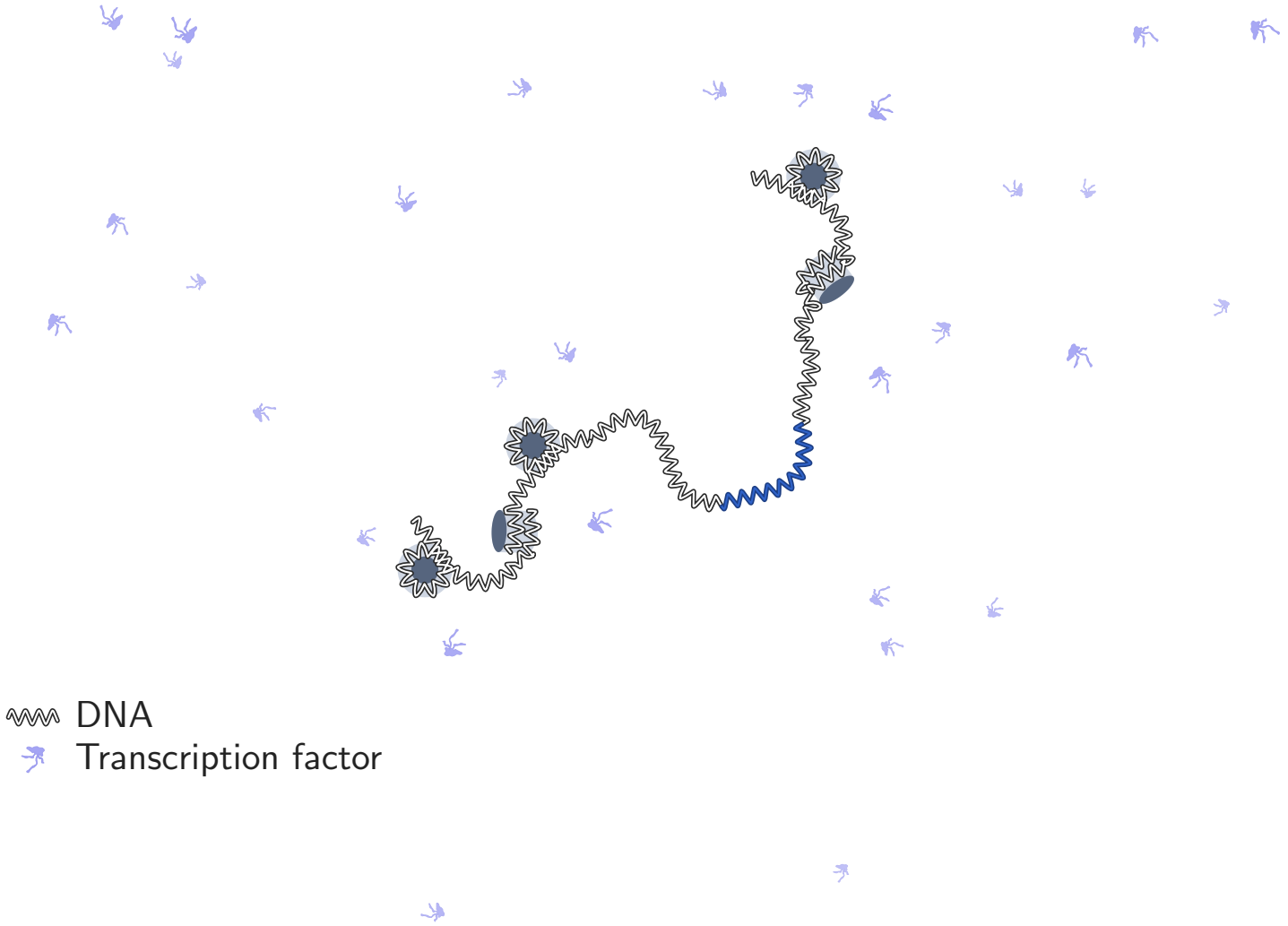
scene6$_3$

... but transcriptional control is noisy



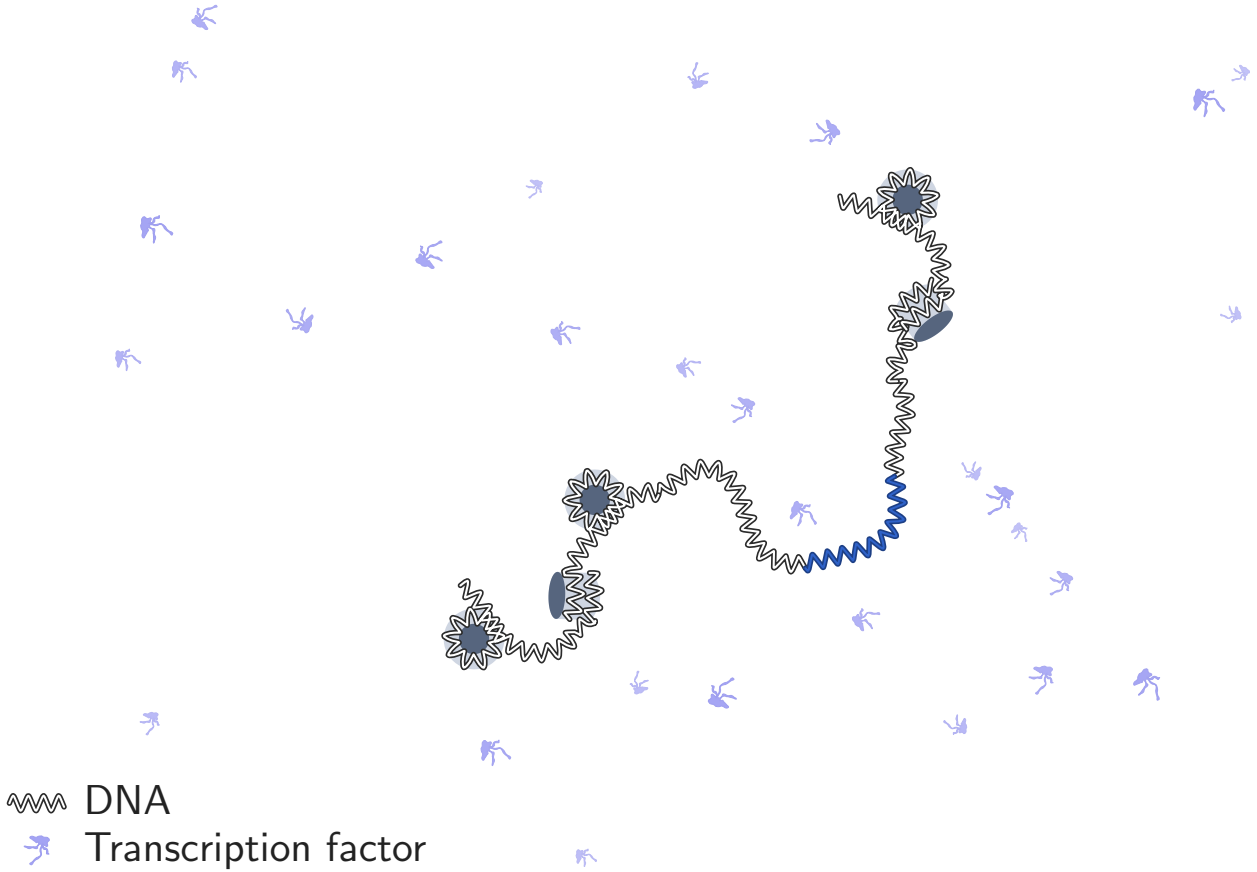
scene6$_4$

... but transcriptional control is noisy



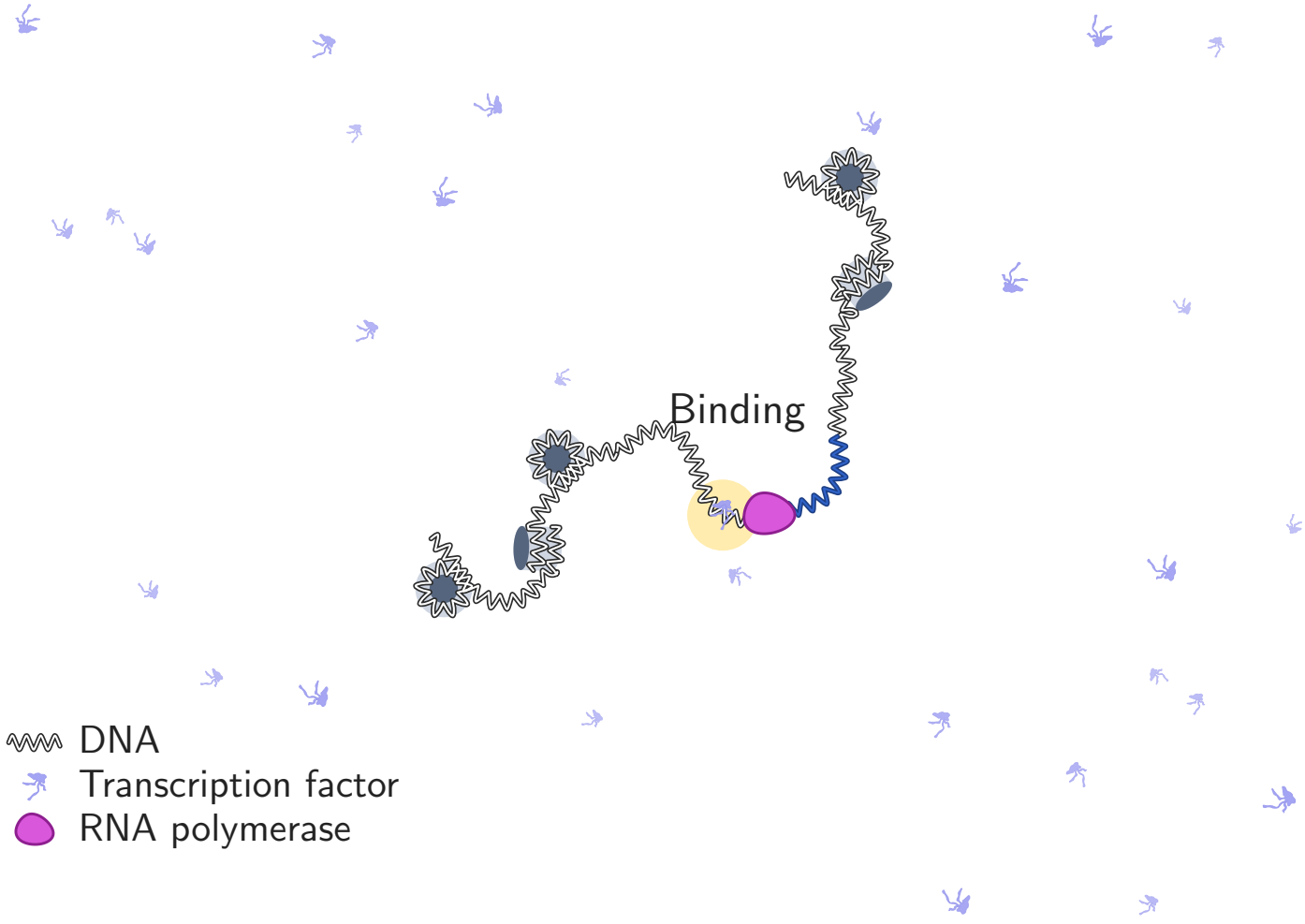
scene6_5

... but transcriptional control is noisy



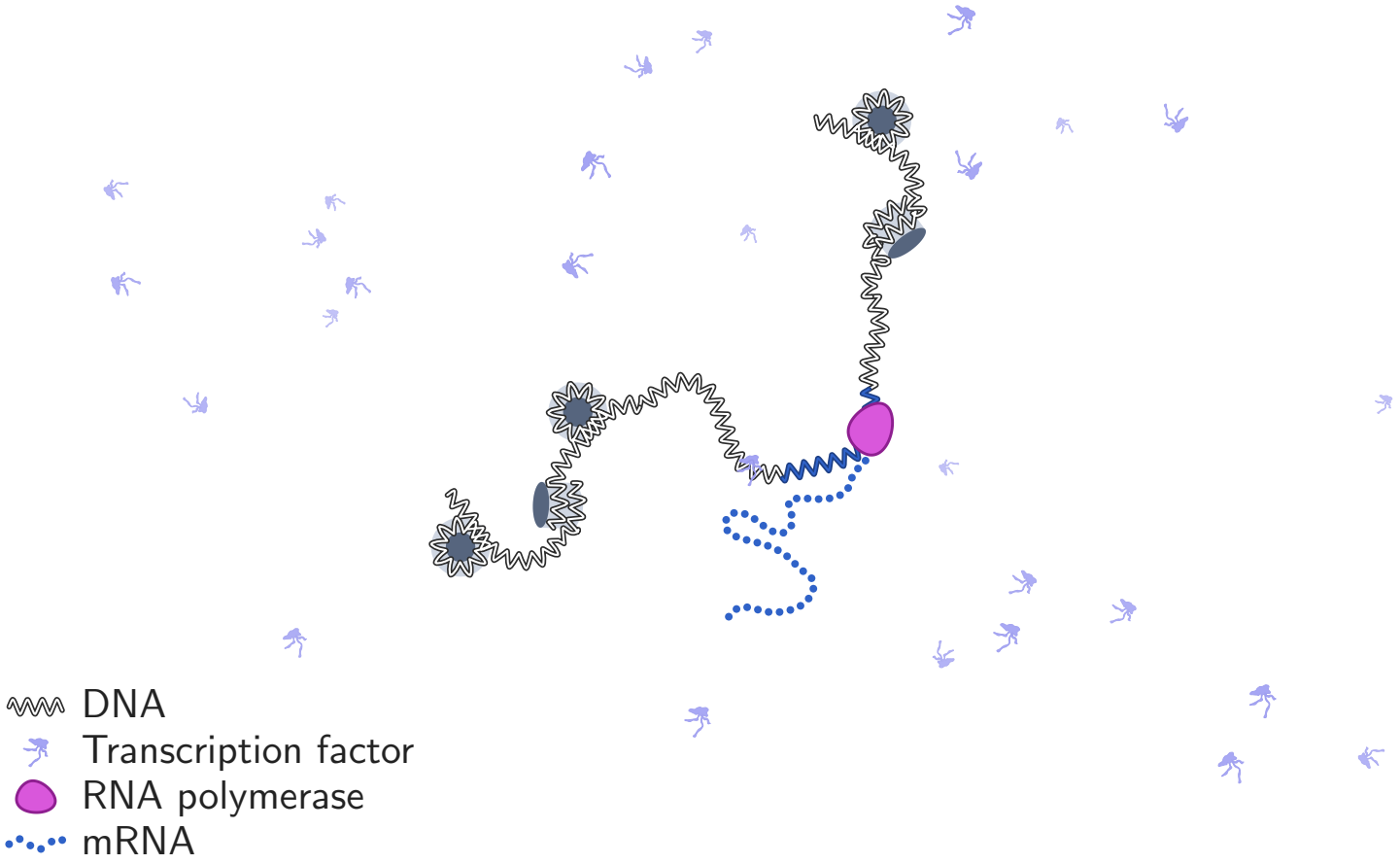
scene6_6

... but transcriptional control is noisy



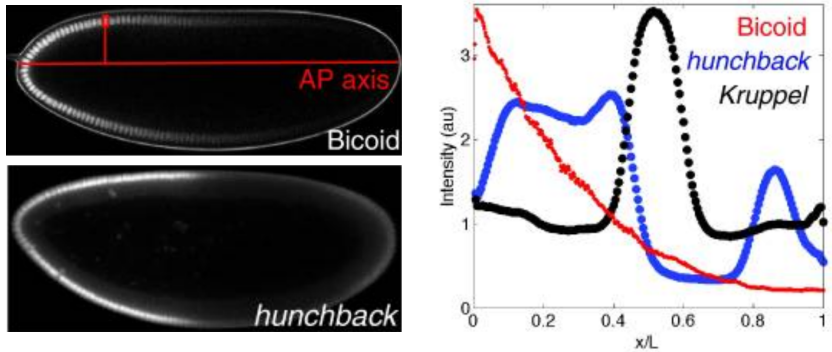
scene6$_7$

... but transcriptional control is noisy



scene6$_8$

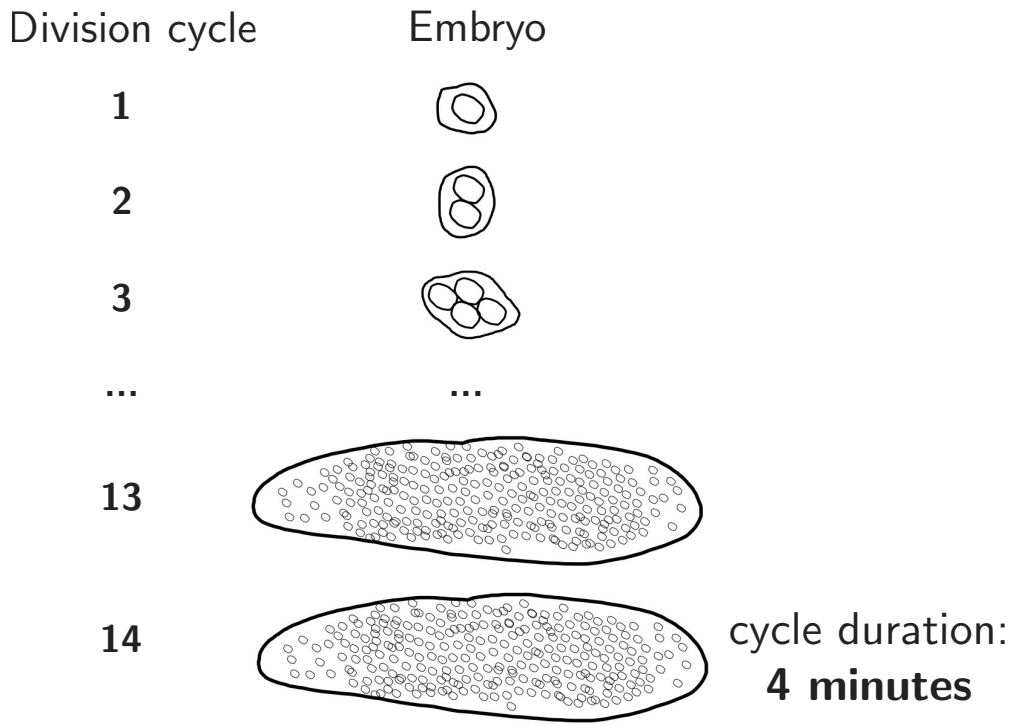
Sensitive response to shallow concentration gradients



Grimm et al., *Development,* 2010

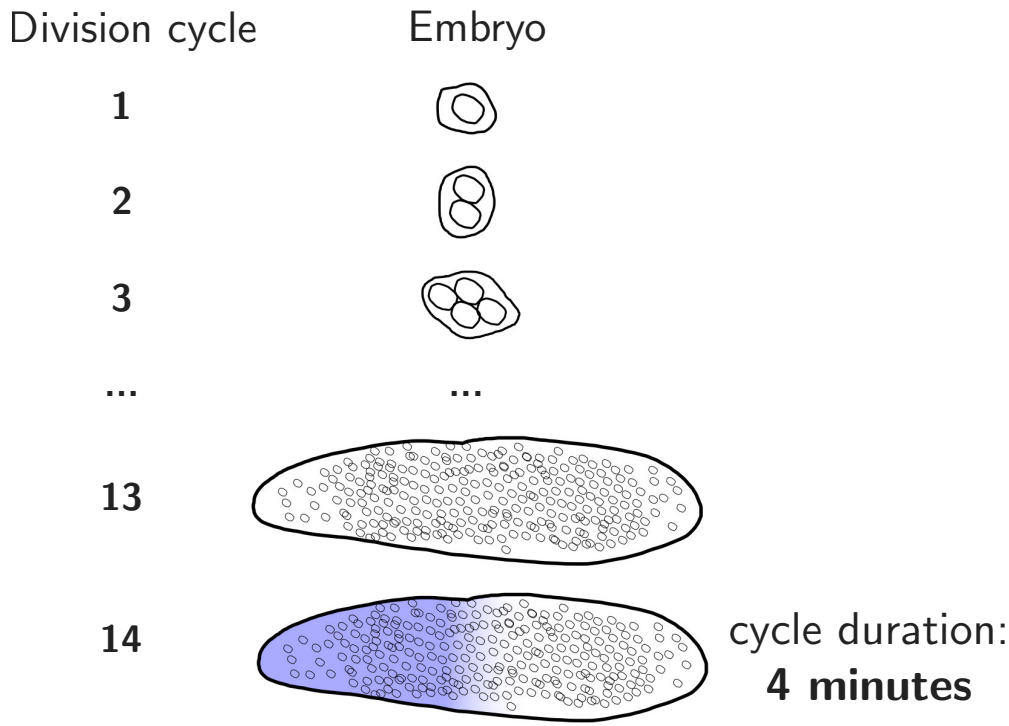
scene7

Precise sensing on a tight schedule



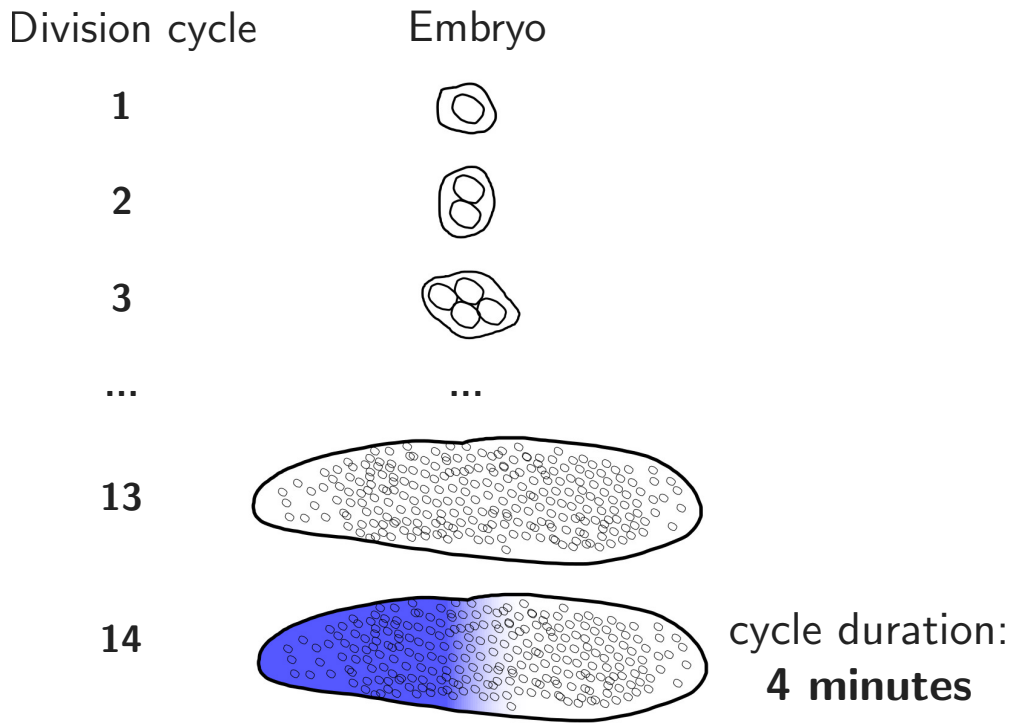
scene8_1

Precise sensing on a tight schedule



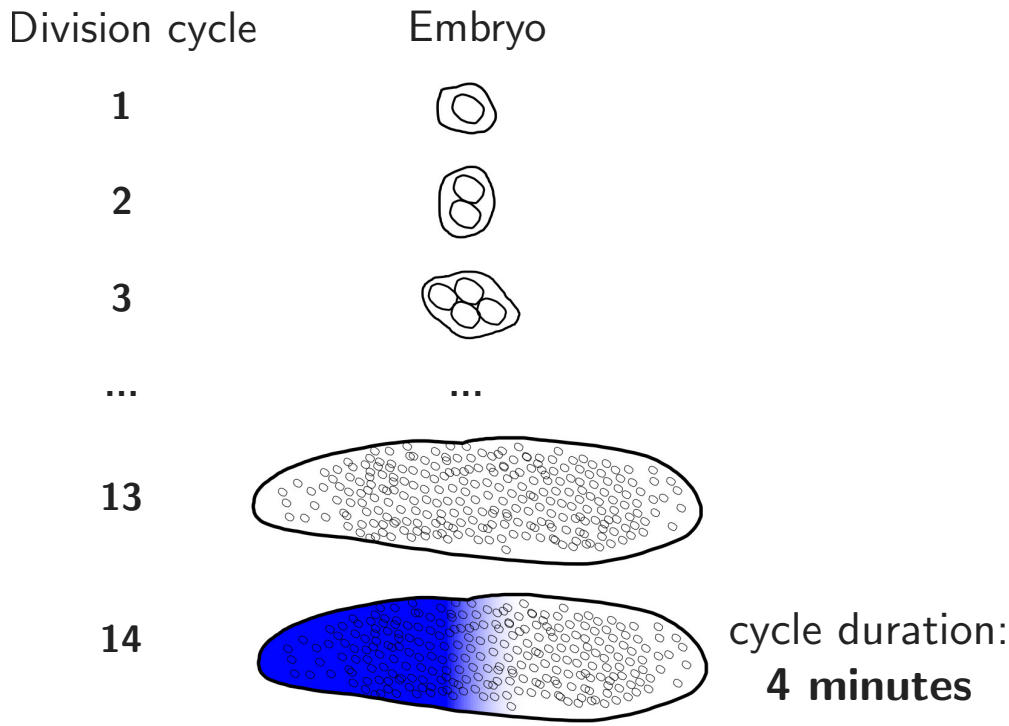
scene8_2

Precise sensing on a tight schedule



scene8_3

Precise sensing on a tight schedule



scene8_4

Understanding the regulatory precision of promoters

Goals
How to quantify precision, to evaluate/compare promoter models?
Focusing on the promoter: which architectures are optimal?
How are these optimal architectures shaped by their constraints?

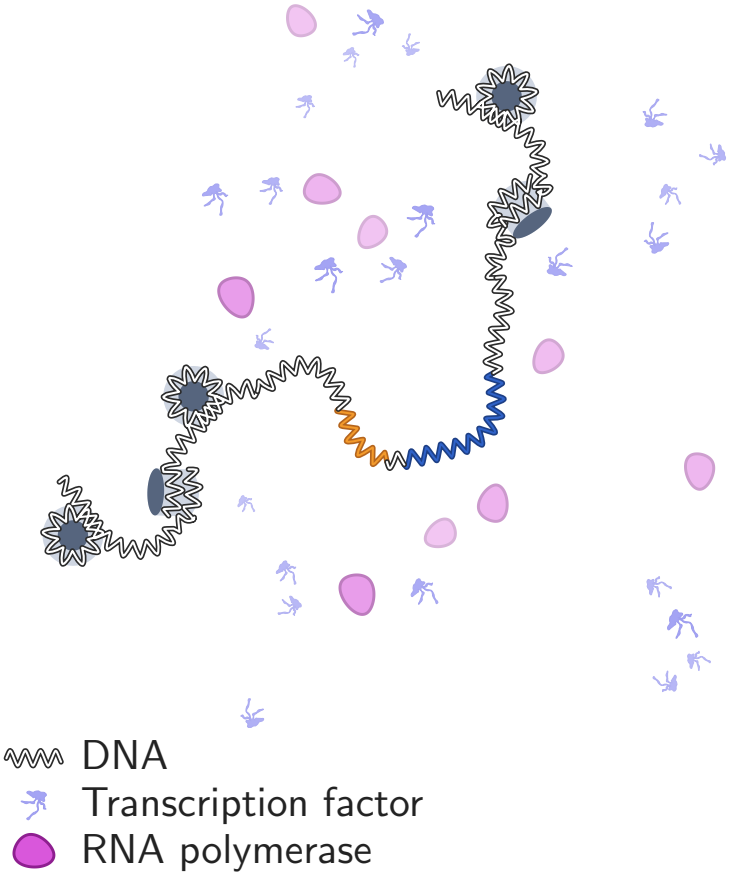
scene9

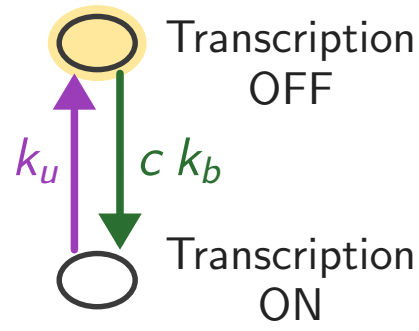
Understanding the regulatory precision of promoters

Approach
Defining the measure of precision
Constrained optimization

scene10

A simple model of transcriptional control

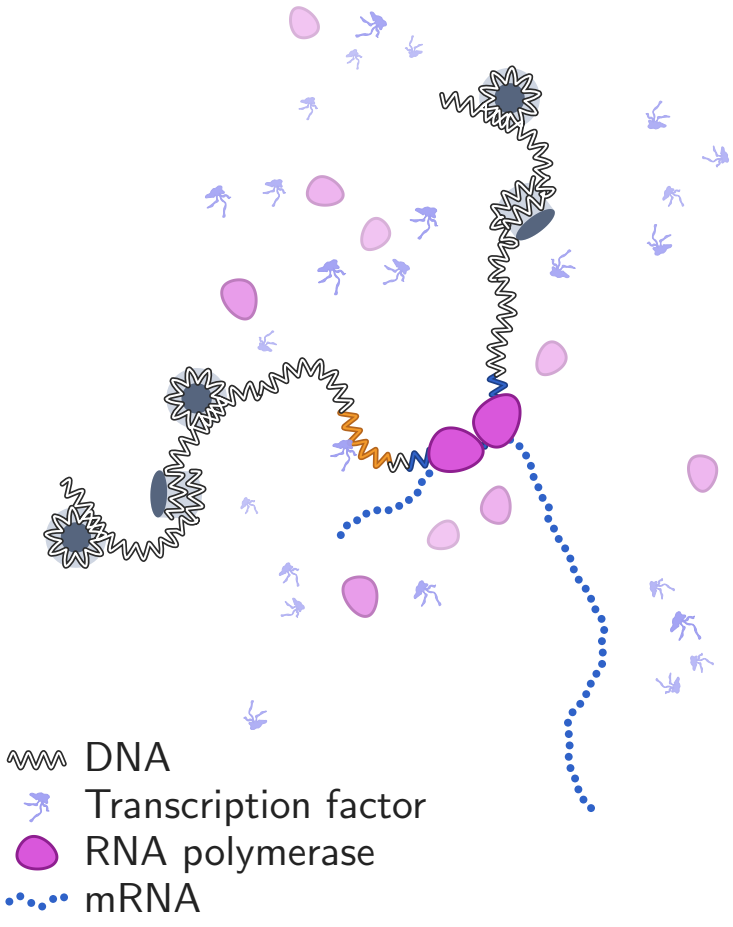


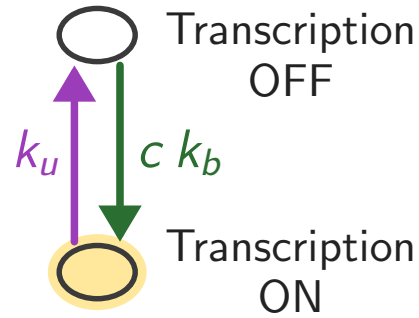




scene11_1

A simple model of transcriptional control

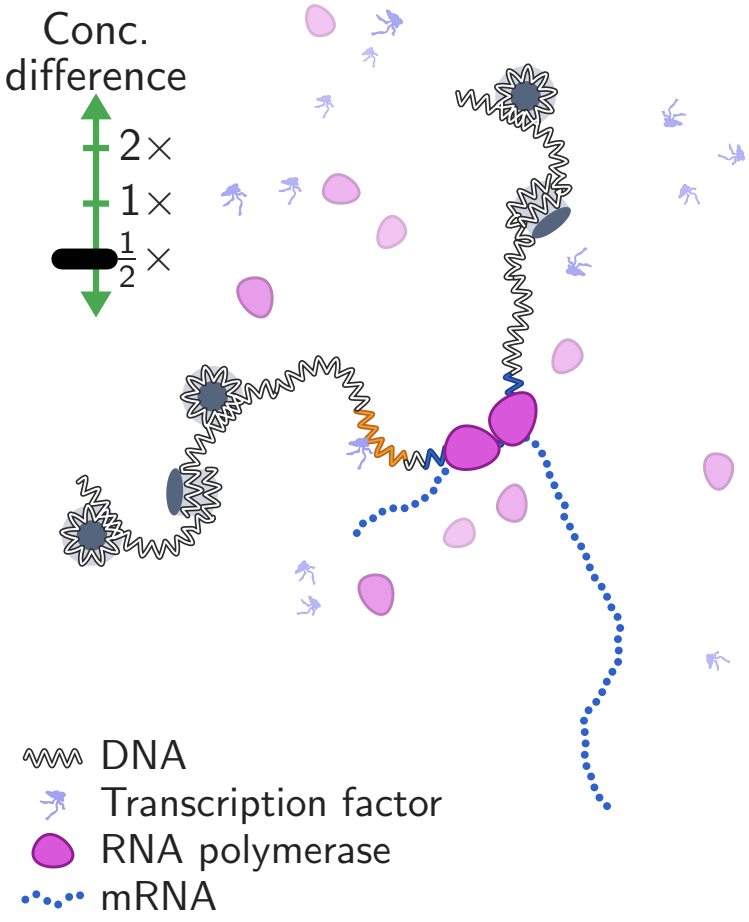


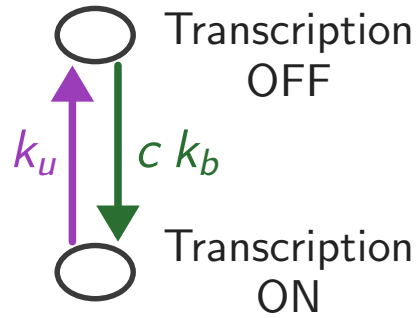


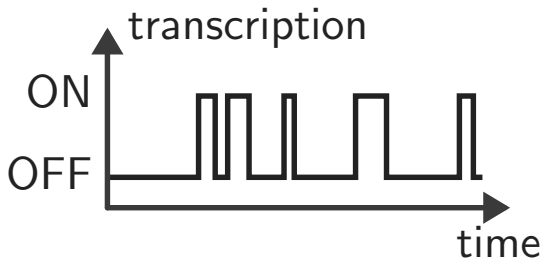


scene11_2

A simple model of transcriptional control

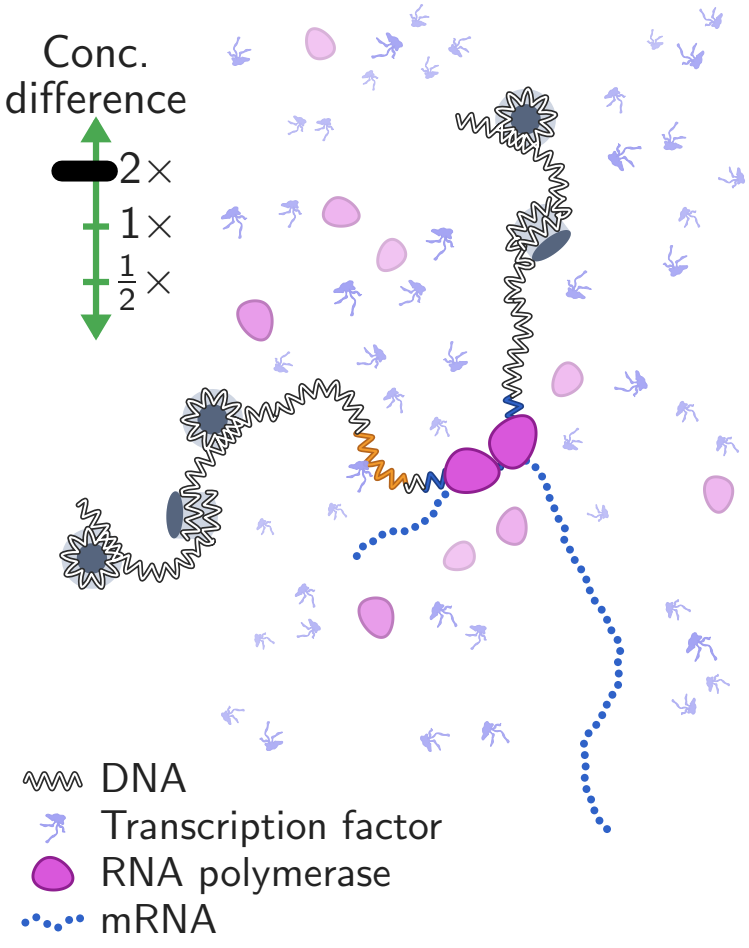


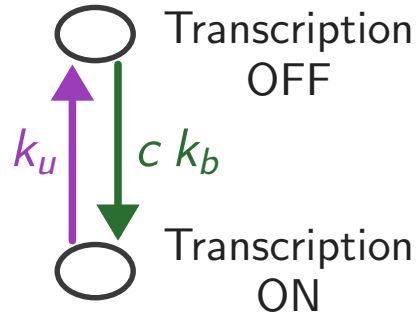


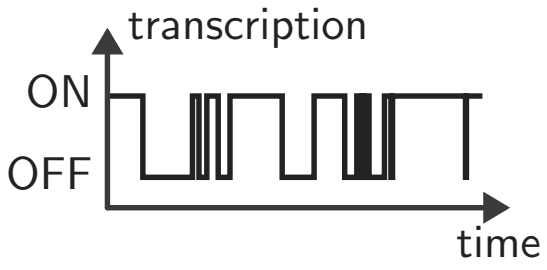


scene11_3

A simple model of transcriptional control







scene11_4

Sequential hypothesis testing

$$P(\text{} \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

$$\frac{P(\text{  } \mid c_{\text{low}})}{P(\text{  } \mid c_{\text{high}})} > 1$$

scene12_1

Sequential hypothesis testing

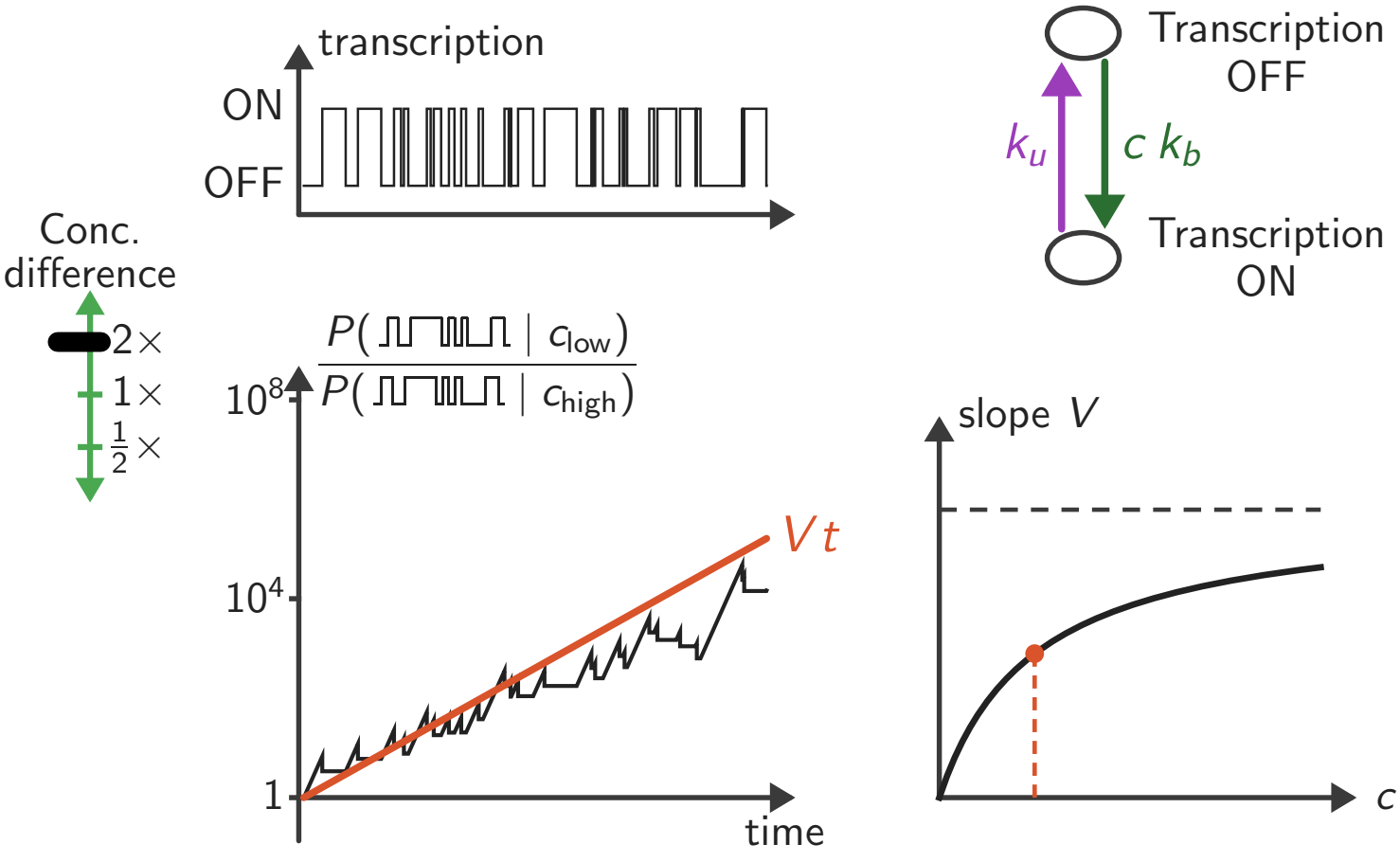
$$P(\text{} \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

$$R(t) = \frac{P(\text{  } \mid c_{\text{low}})}{P(\text{} \mid c_{\text{high}})} > 1$$

$$\log R(t) = V t + \sqrt{2D}\,\xi$$

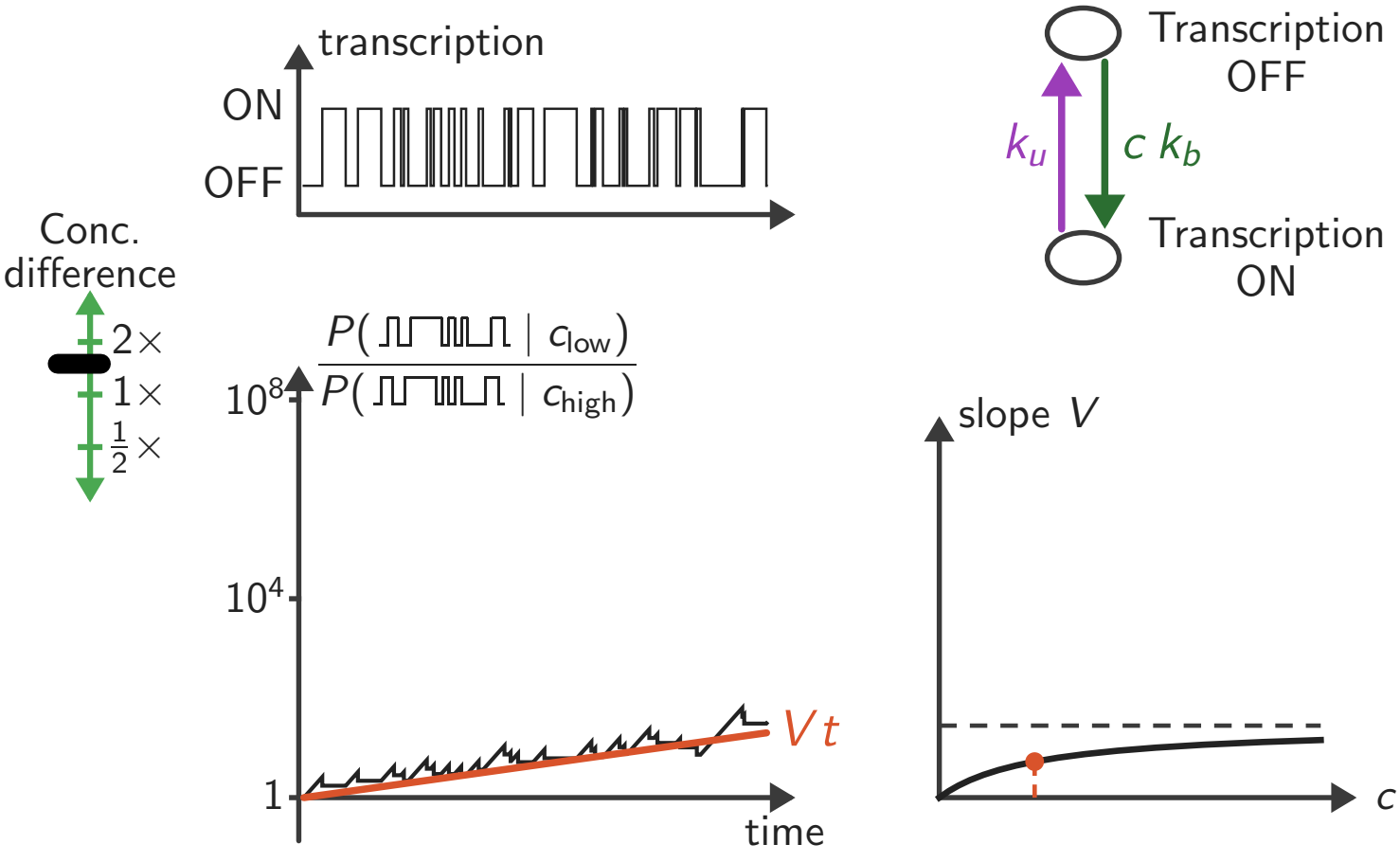
scene12_2

Sequential hypothesis testing



scene13_1

Sequential hypothesis testing



scene13_2

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

scene14_1

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

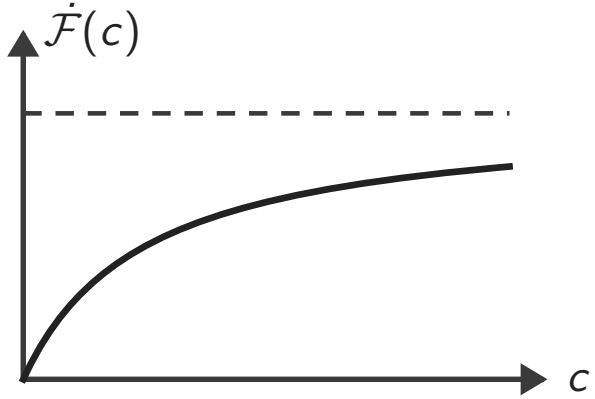
scene14_2

Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$



$c\,k_b$: diffusion-limited arrival of molecules

$k_b = 4aD$

scene14_3

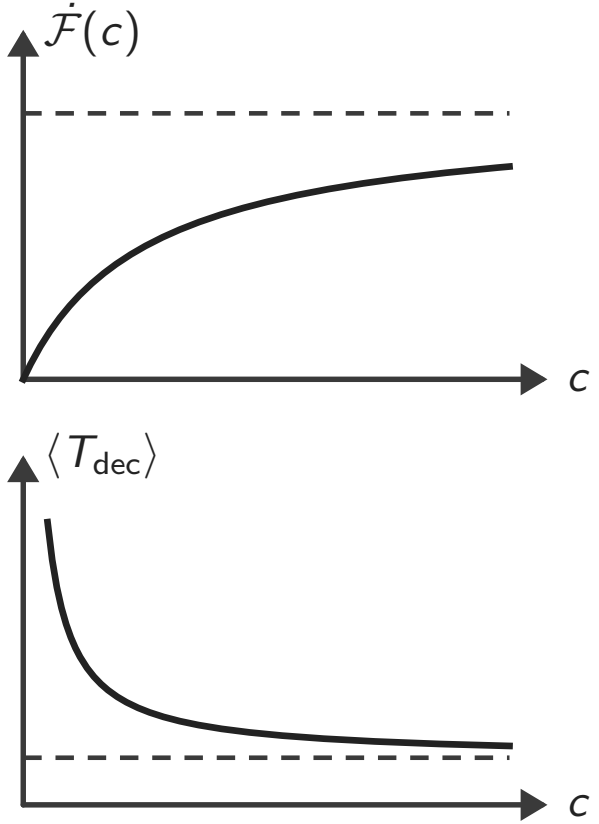
Fisher information rate and decision time

$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}_{\text{Fisher information rate}}$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty}\frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2}\right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$

$$\langle T_{\text{dec}}\rangle = \frac{K}{V}\tanh\left(\frac{K}{2}\right)$$

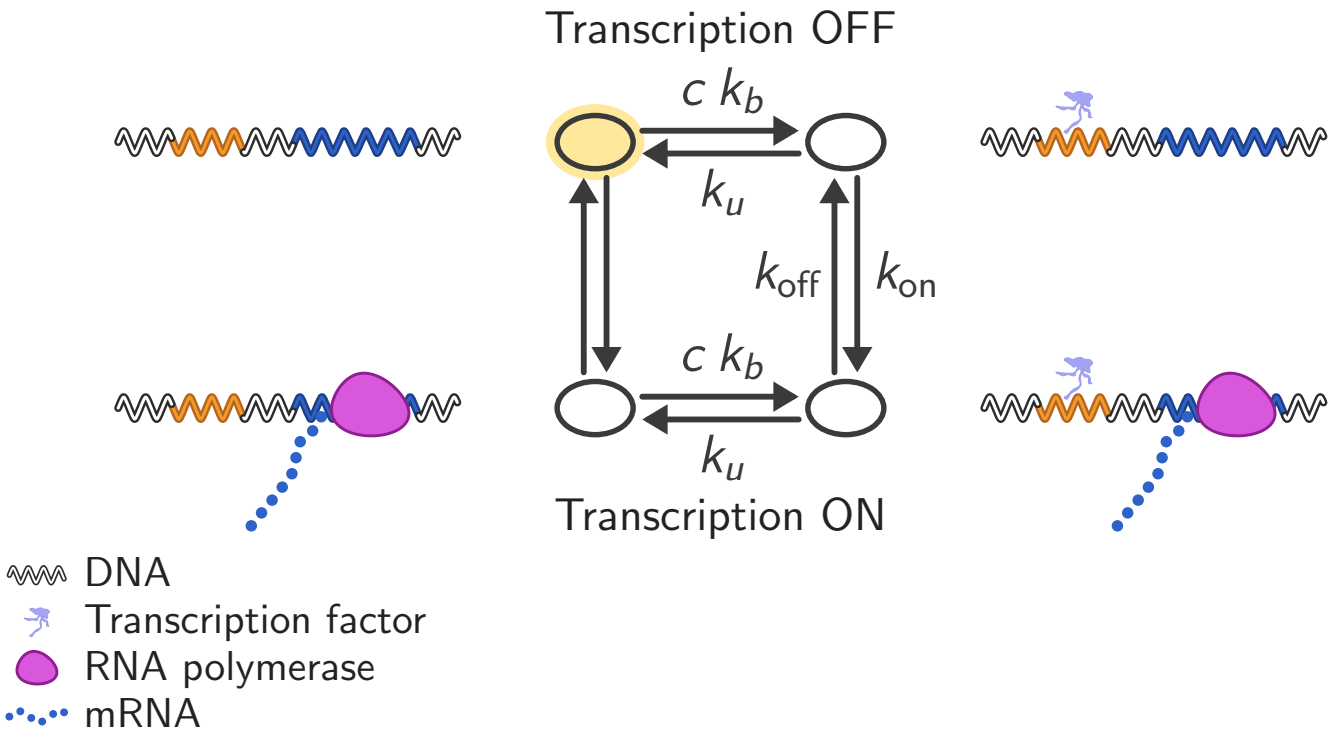


$c\,k_b$: diffusion-limited arrival of molecules

$k_b = 4aD$

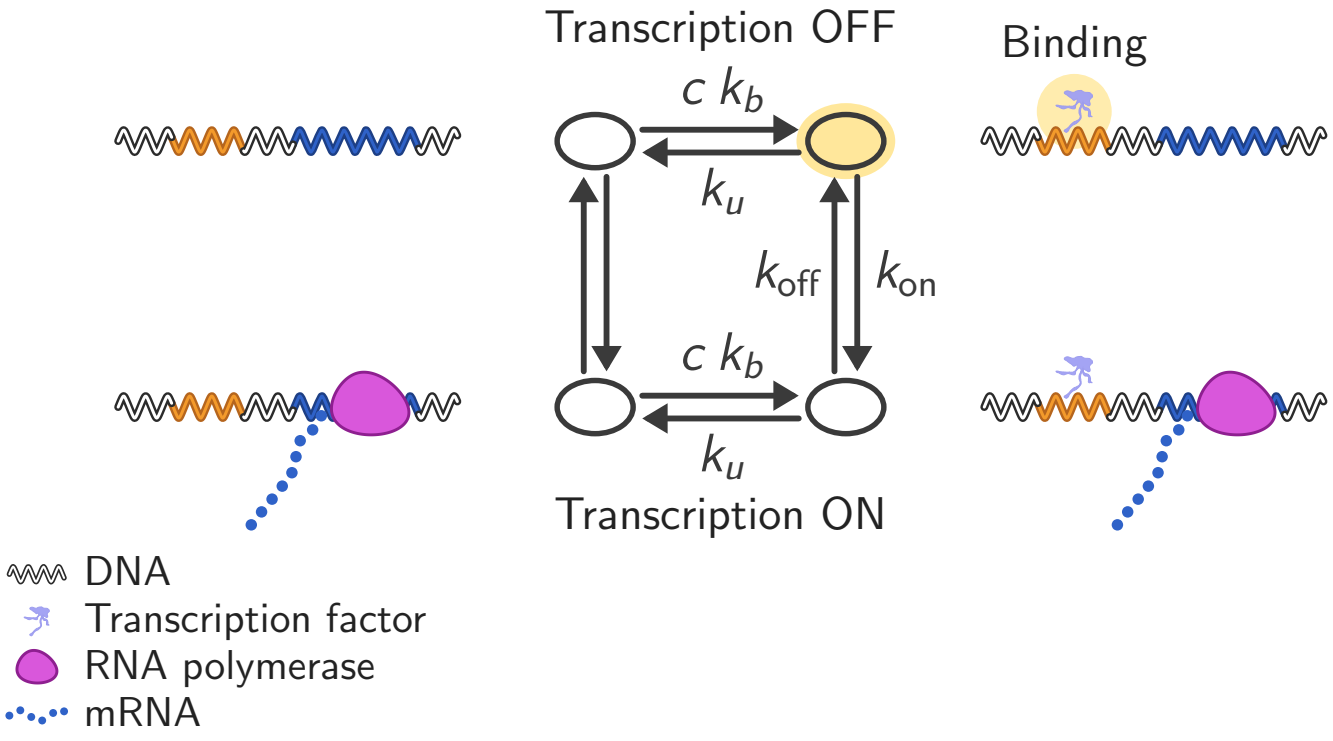
scene14$_4$

Modeling time-limited activation



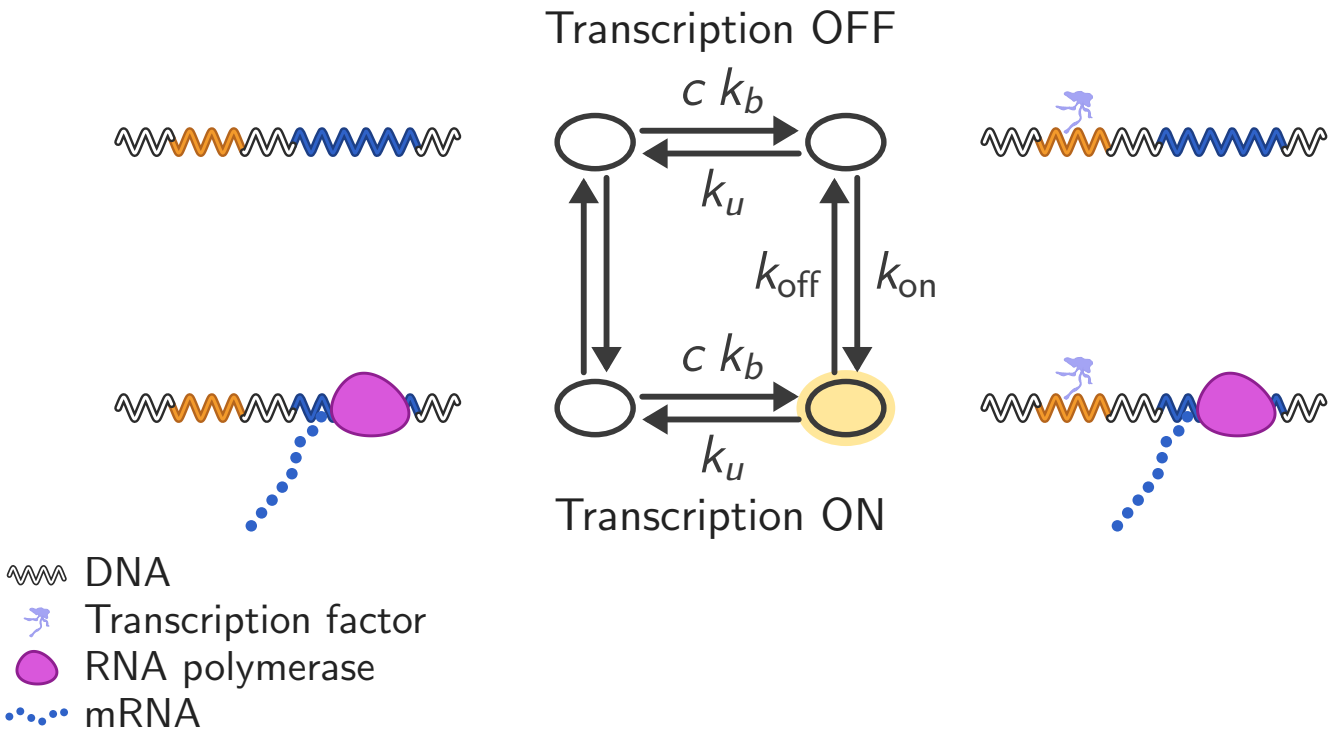
scene15_1

Modeling time-limited activation



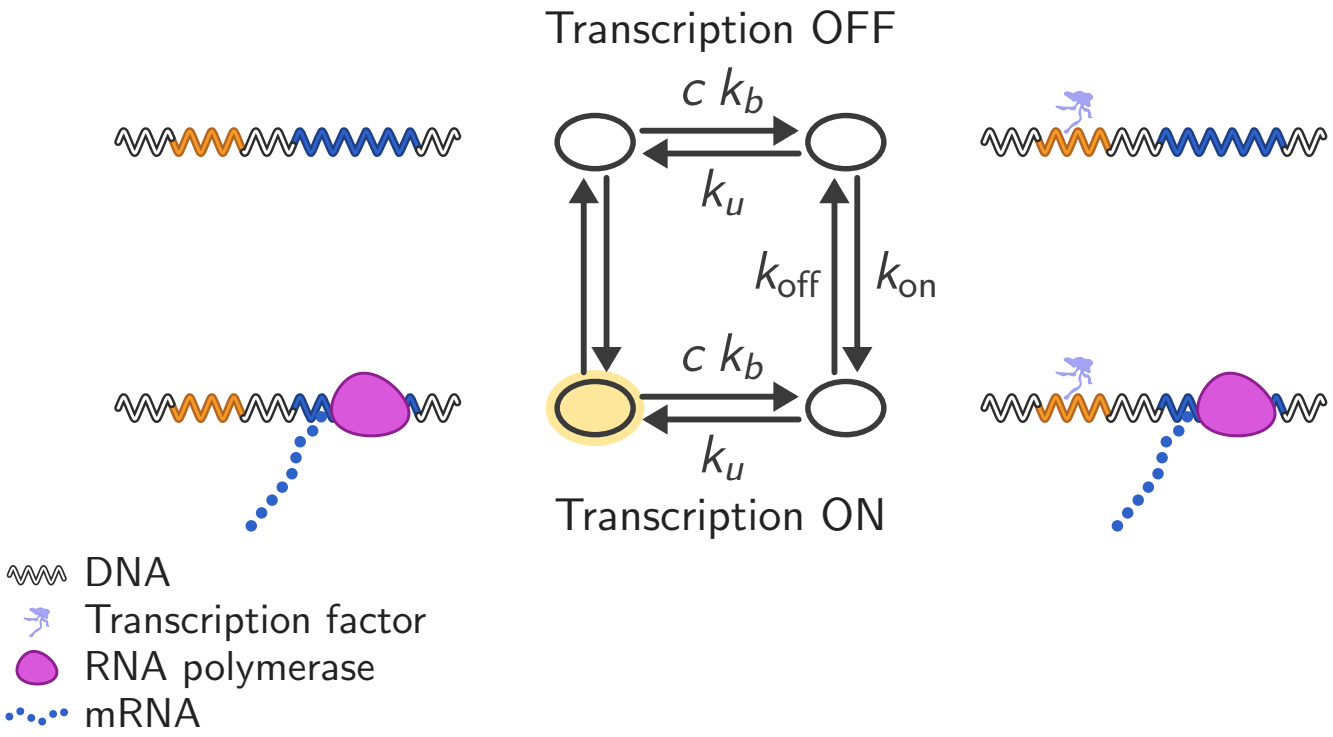
scene15_2

Modeling time-limited activation



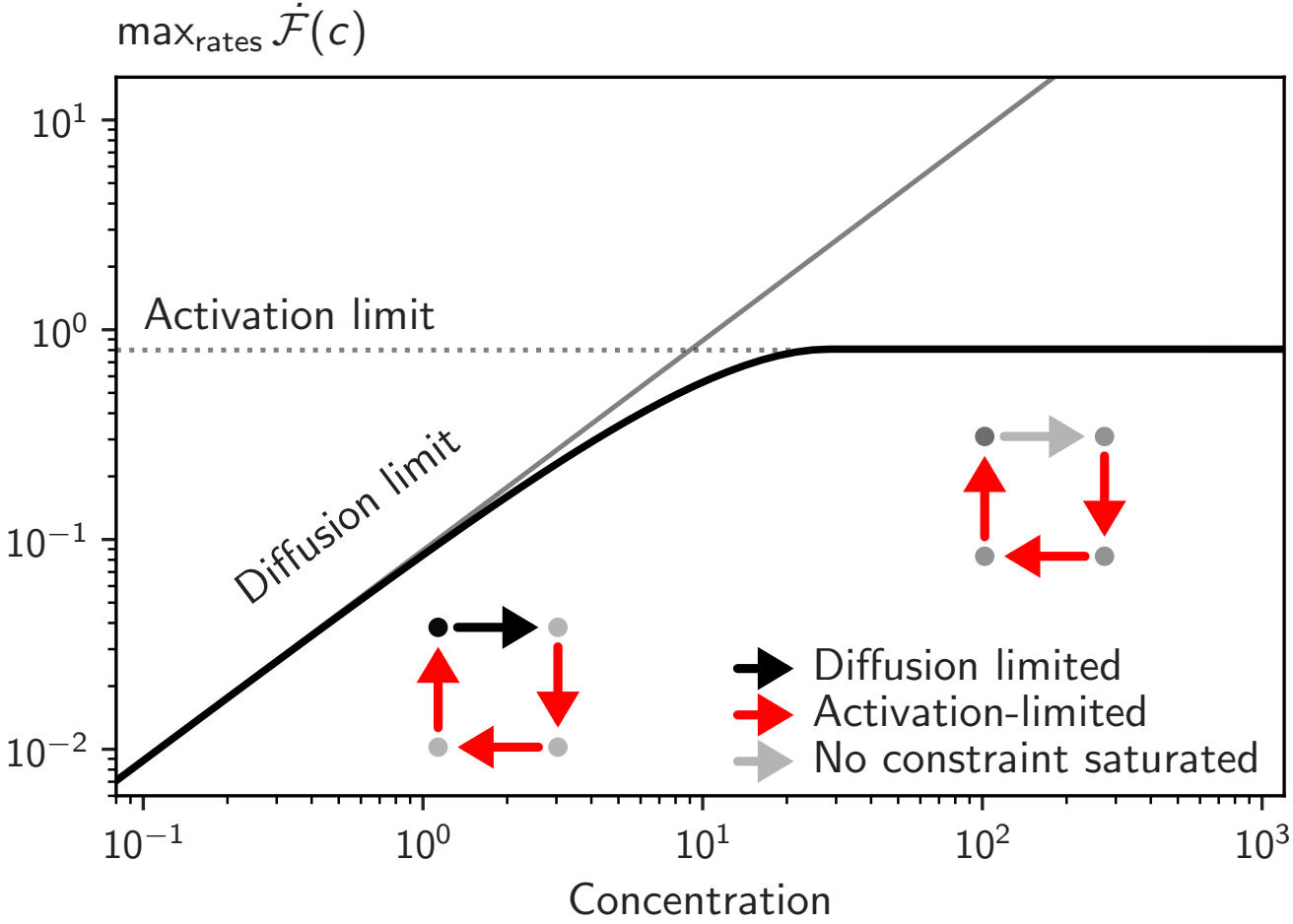
scene15_3

Modeling time-limited activation



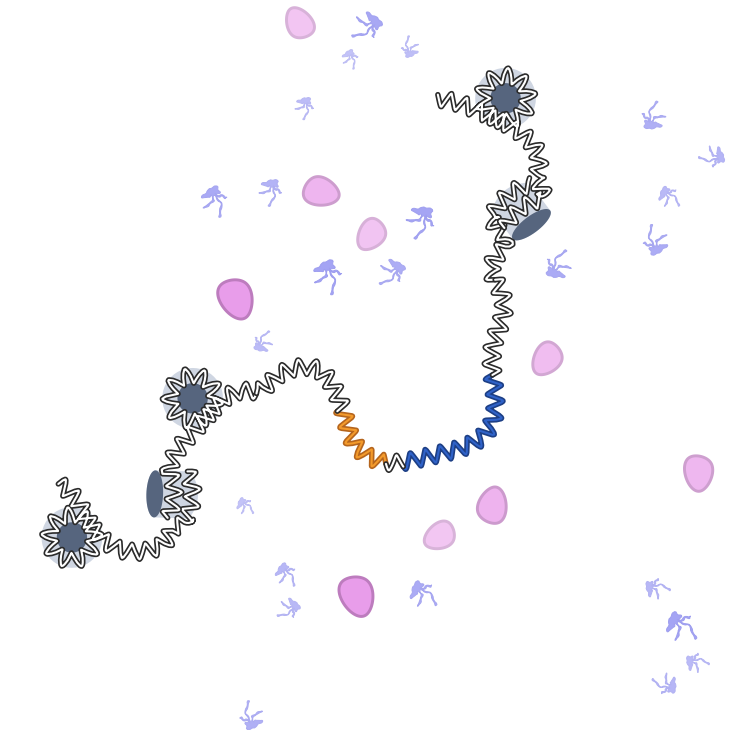
scene15_4

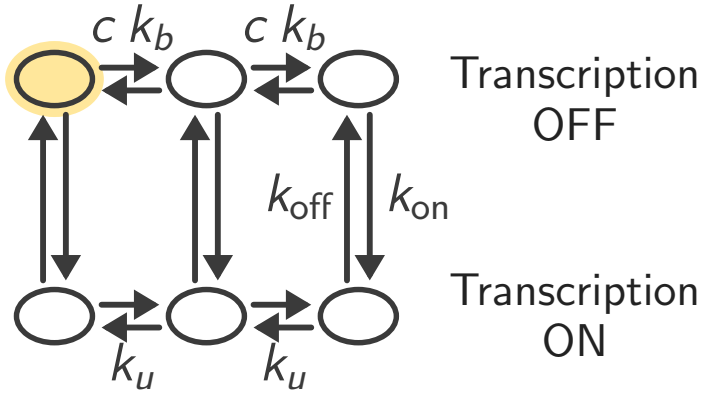
Optimal promoters: two distinct regimes



scene16

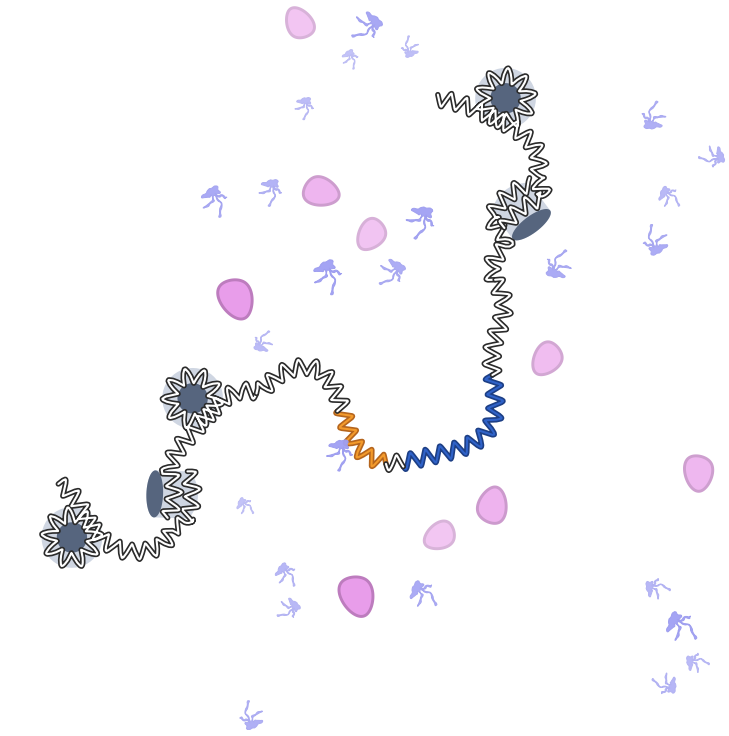
Cooperative binding

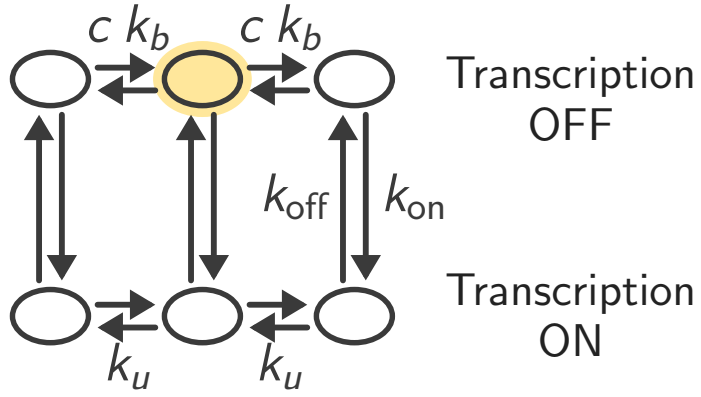




scene17_1

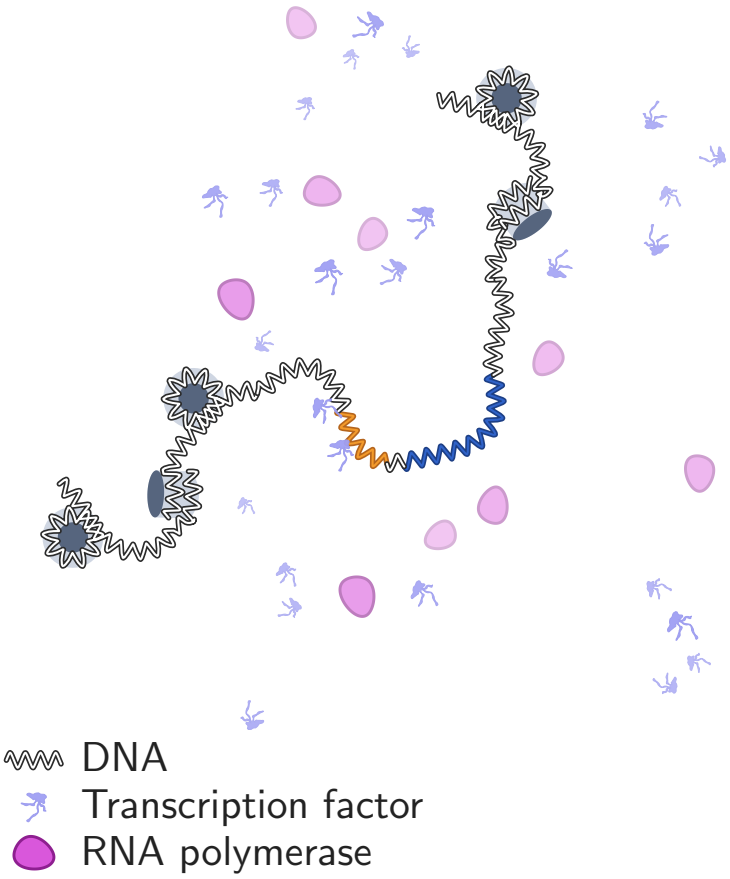
Cooperative binding

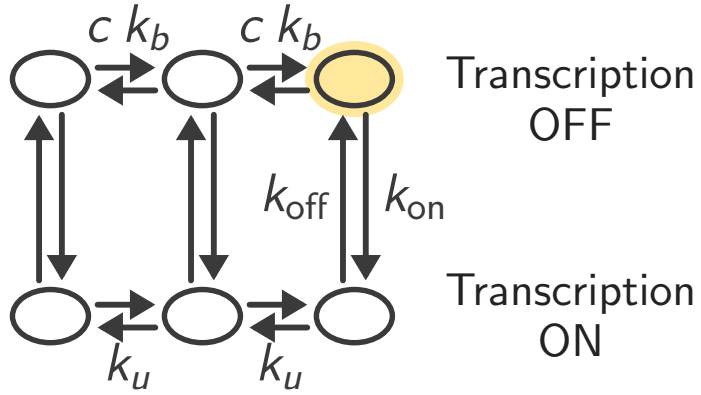




scene17_2

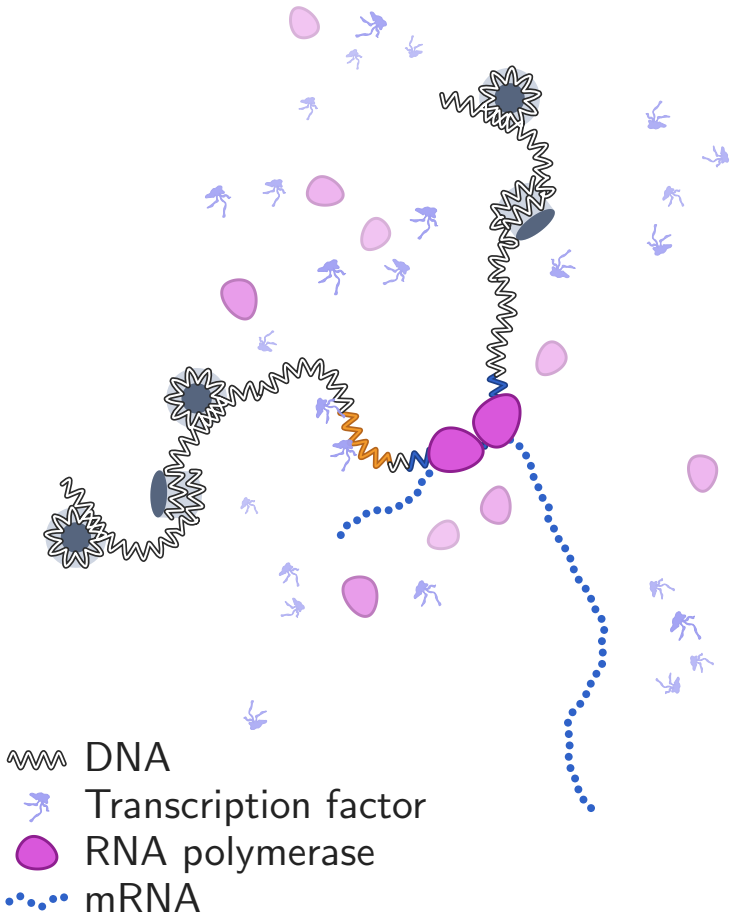
Cooperative binding

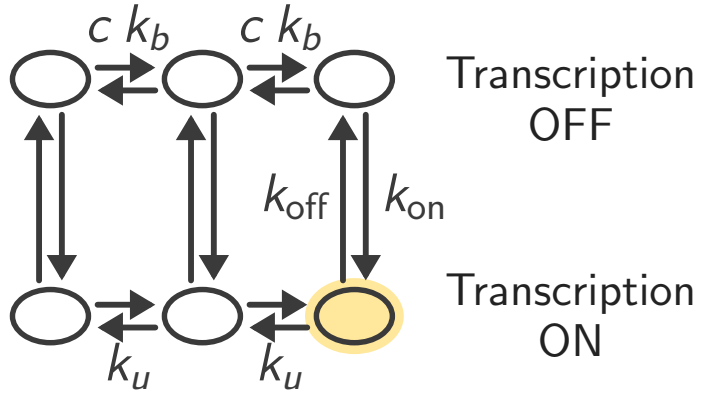




scene17_3

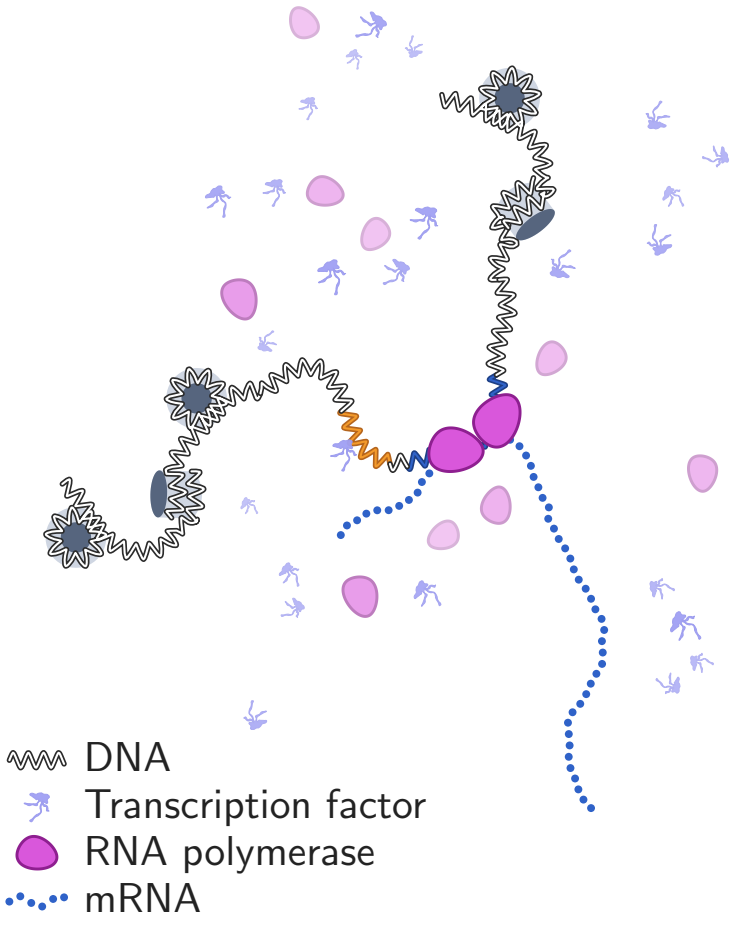
Cooperative binding

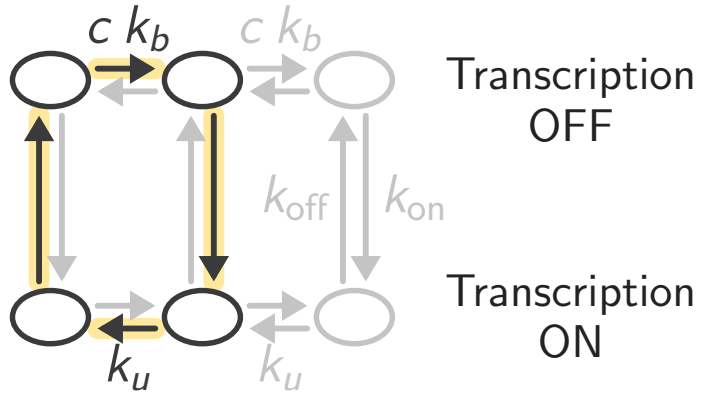


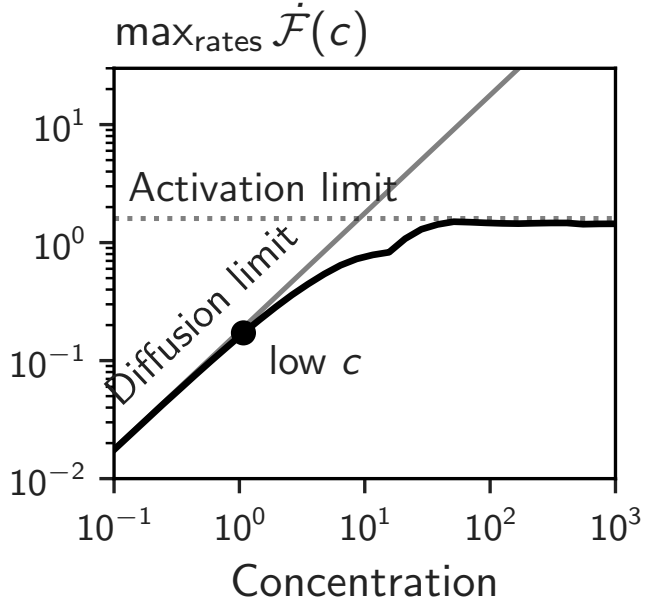


scene17_4

Cooperative binding

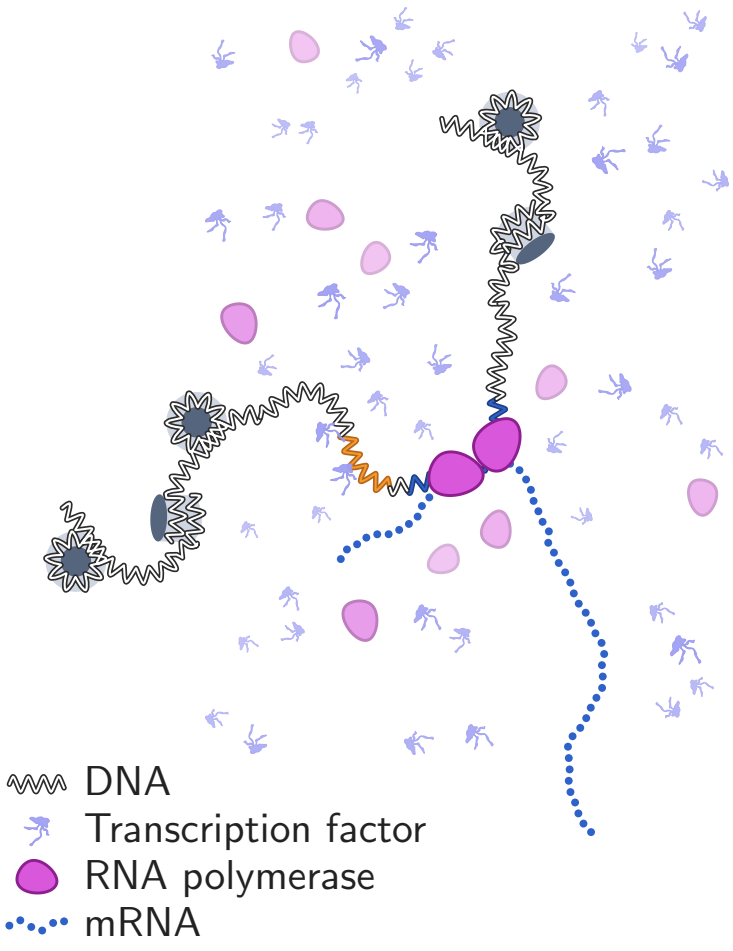


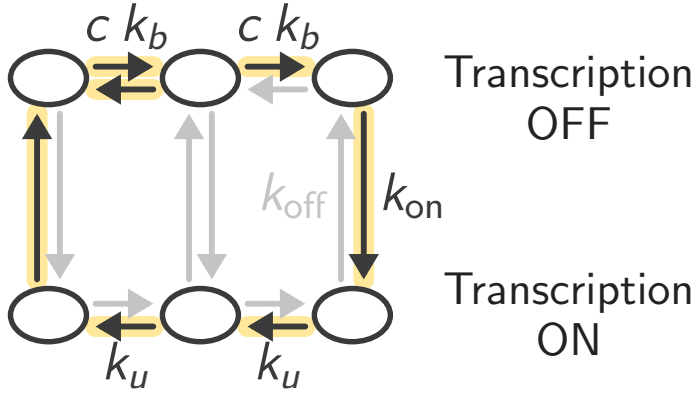


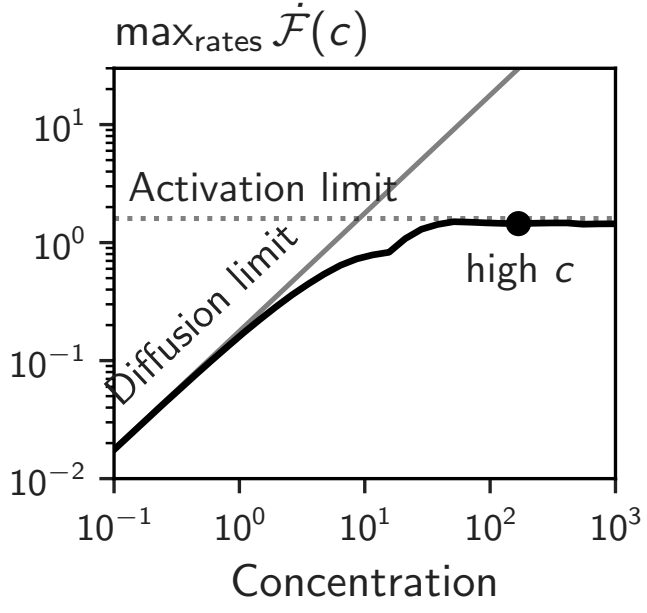


scene17_5

Cooperative binding

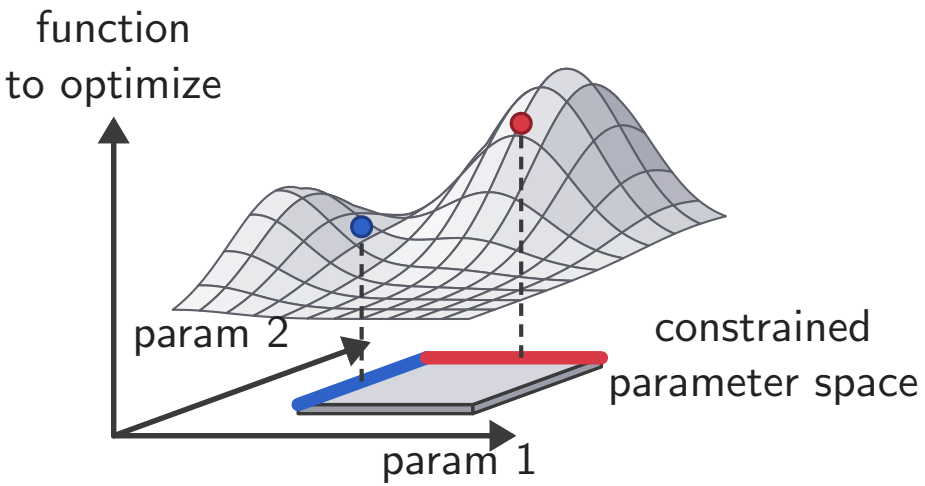




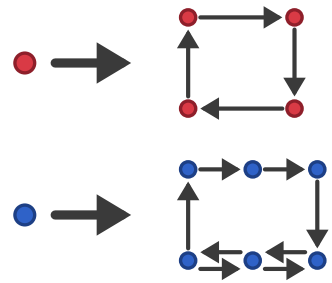


scene17$_6$

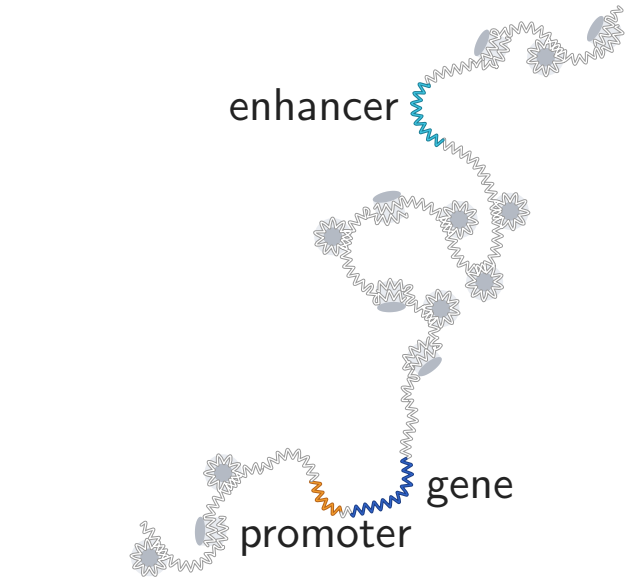
Takeaway



same optimized function
different mechanisms



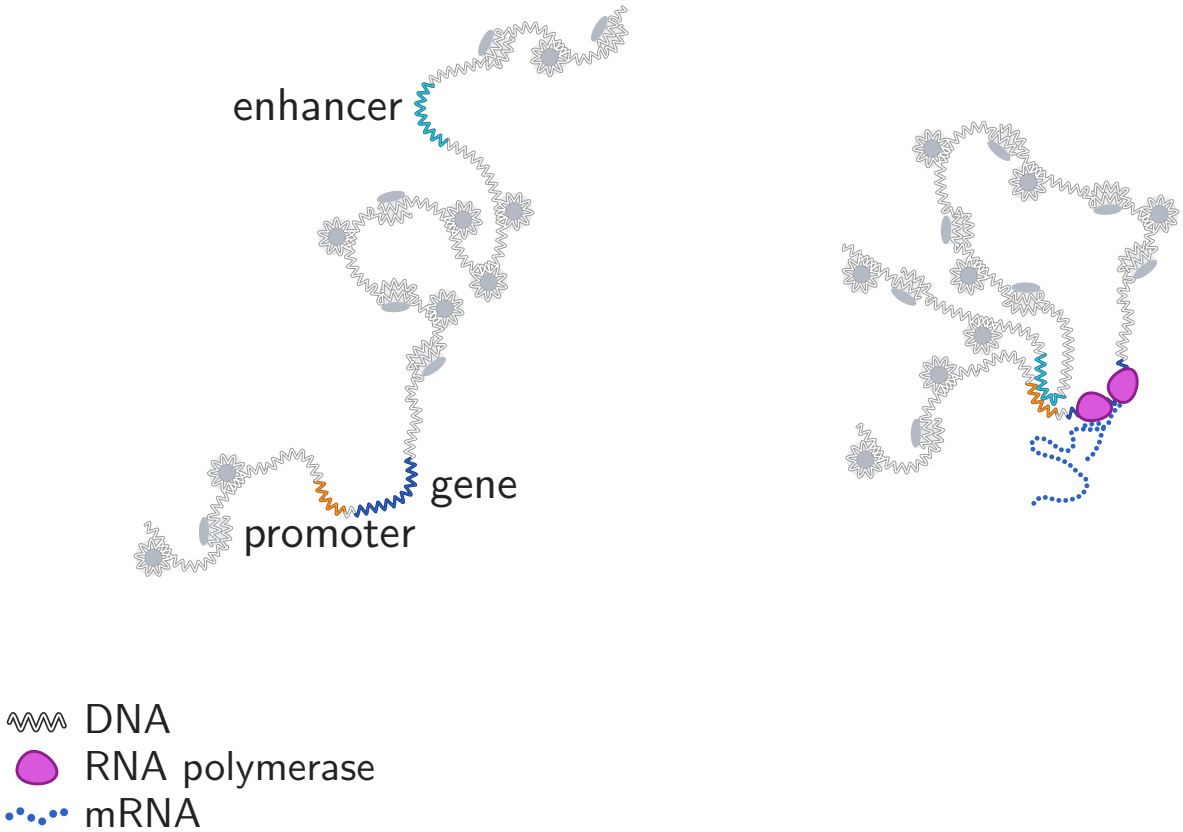
Transcriptional control by distal elements



DNA

scene20_1

Transcriptional control by distal elements

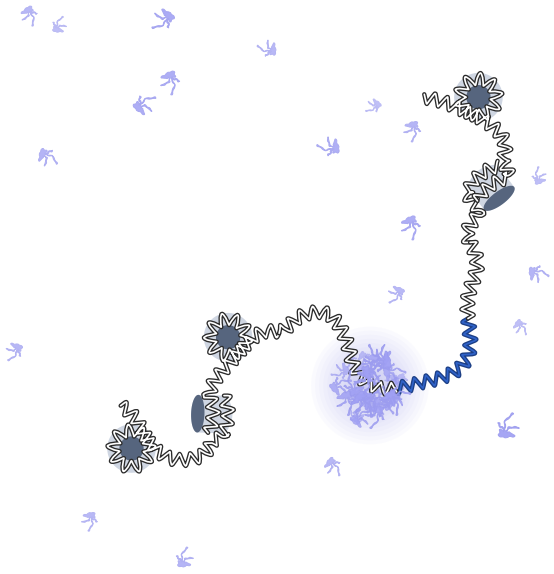


scene20_2

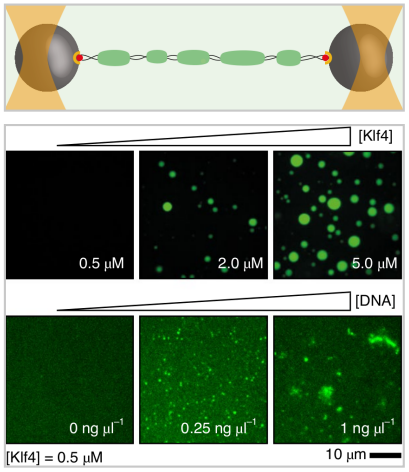
Transcriptional control through liquid phase separation?

scene21$_1$

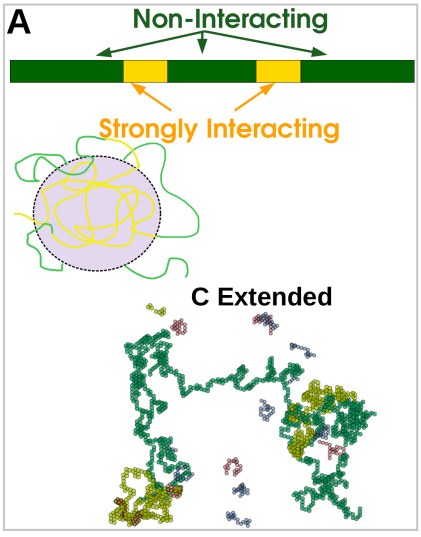
Proteins condense on the surface of DNA



DNA
Transcription factor



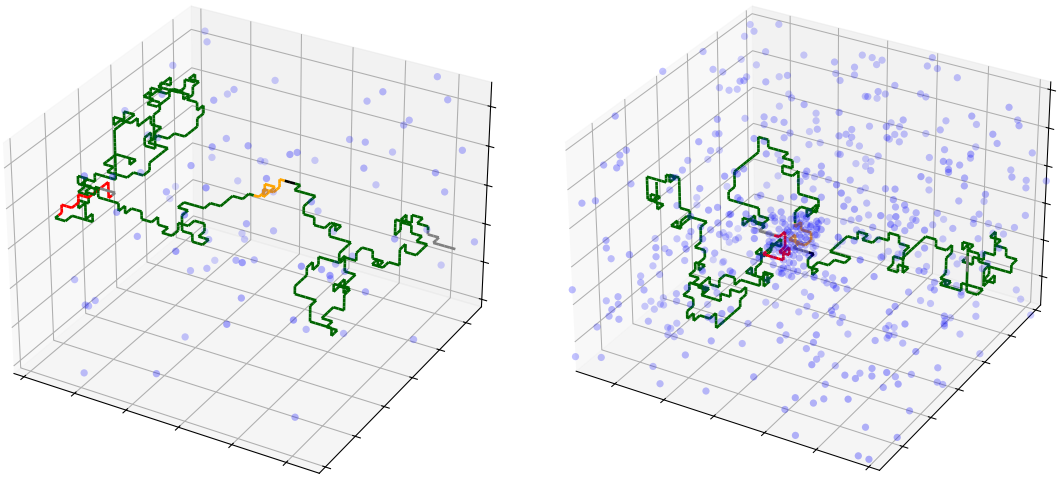
Sequence-dependent surface condensation of a pioneer transcription factor on DNA
Morin et al., 2022

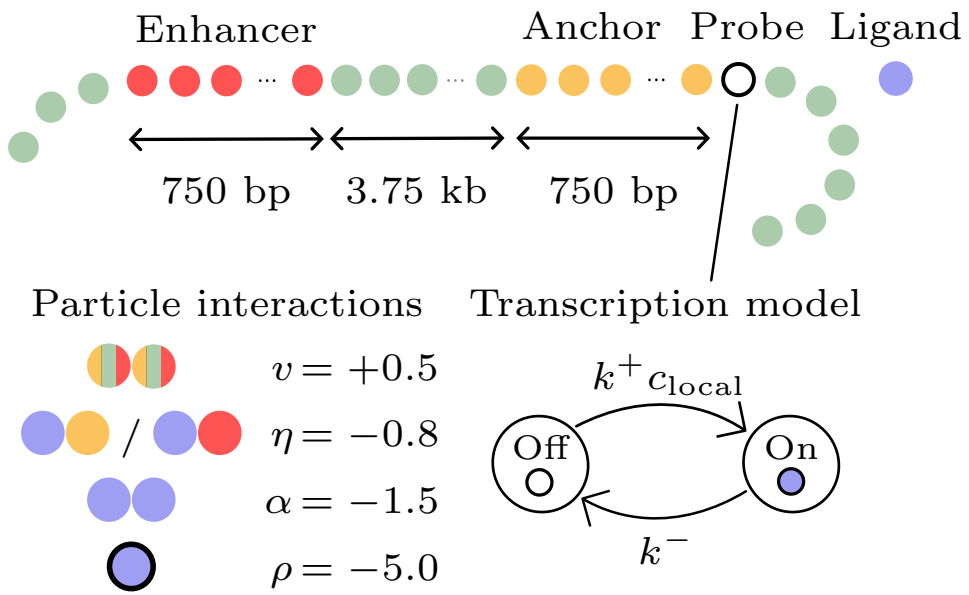


Protein-DNA Co-condensation is Prewetting to a Collapsed Polymer
Rouches & Machta, 2025

scene21$_2$

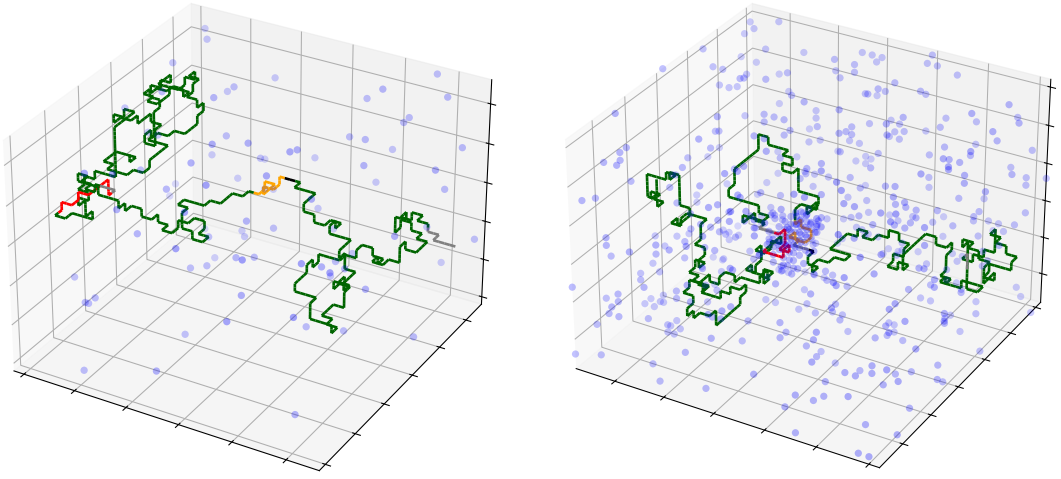
Coupling condensation and polymer dynamics

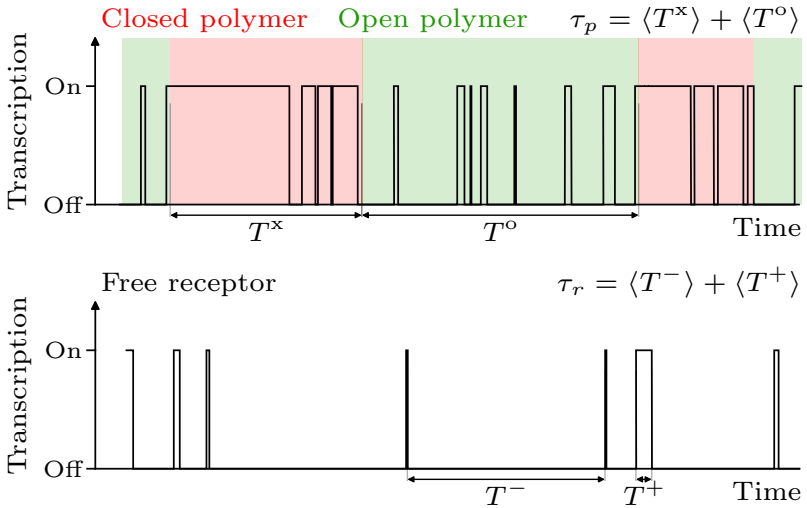


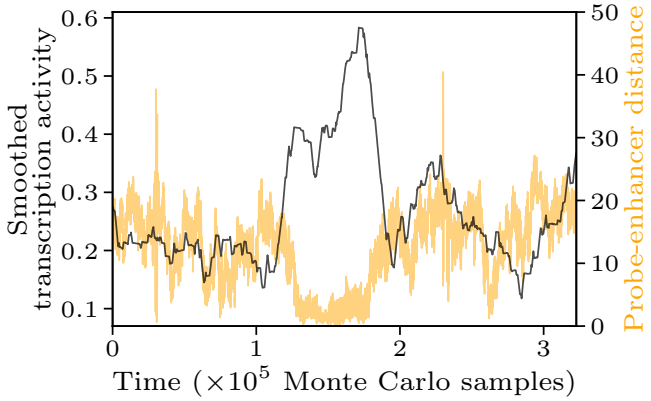


scene22

Coupling condensation and polymer dynamics

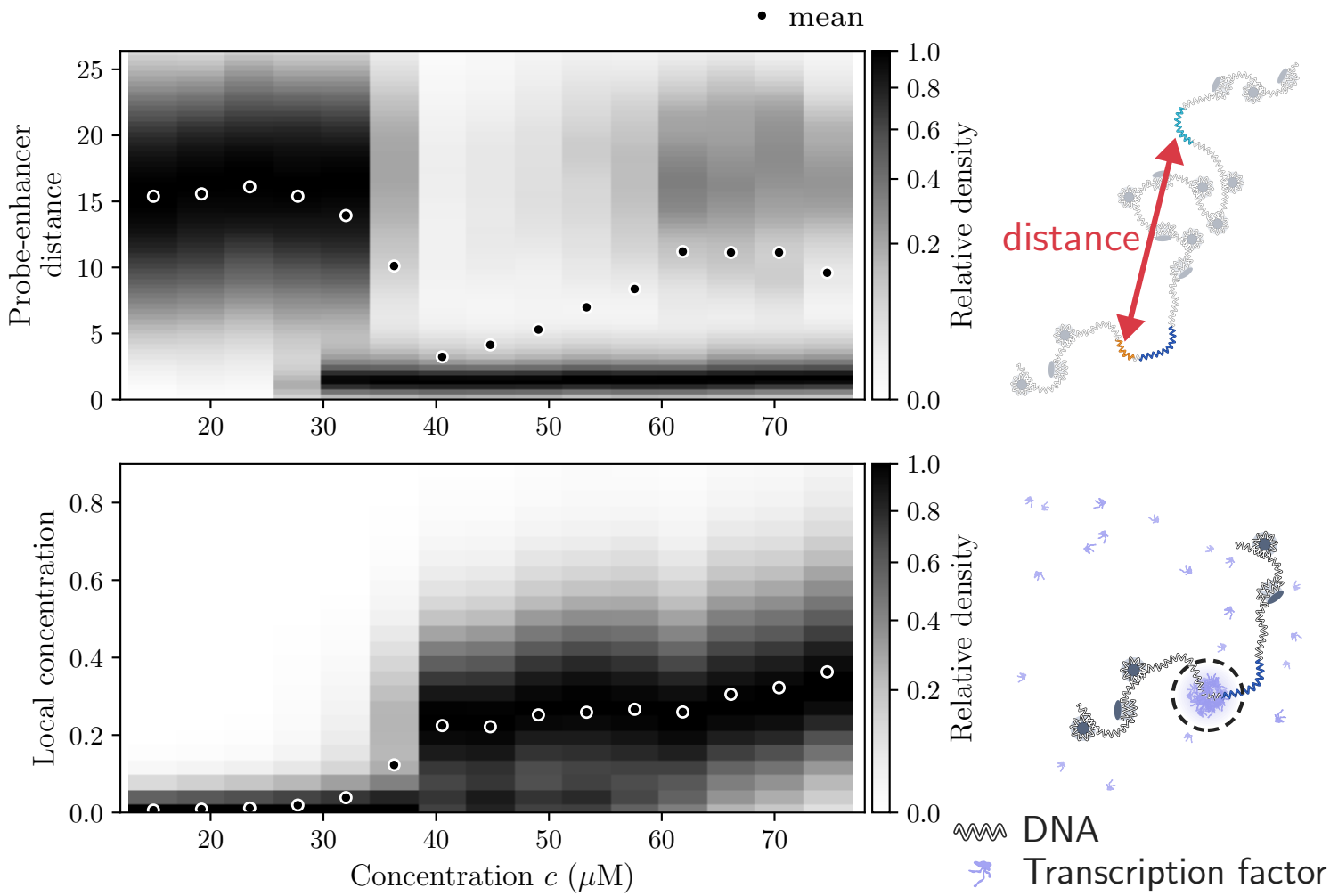






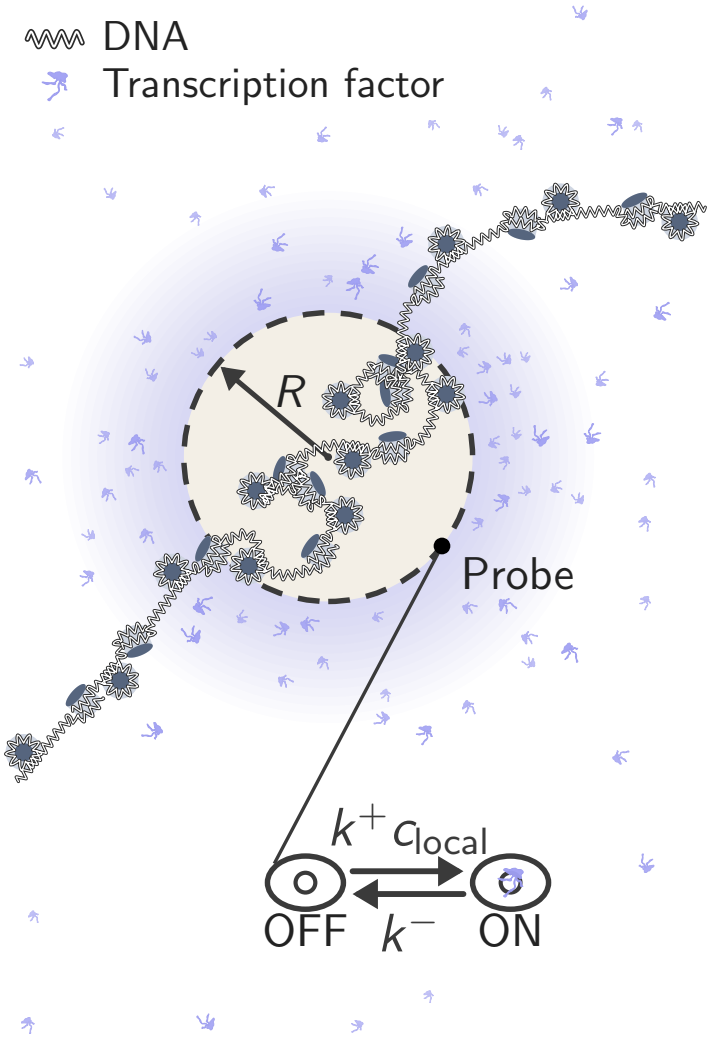
scene23$_2$

Polymer configuration depends sensitively on concentration



scene24

Continuous model of surface condensation



Flory–Huggins potential

$$f(\phi) = \frac{\phi}{\nu} \ln \phi + (1-\phi)\ln(1-\phi) - \chi\phi^2 - \mu\phi$$

Cahn–Hilliard equation

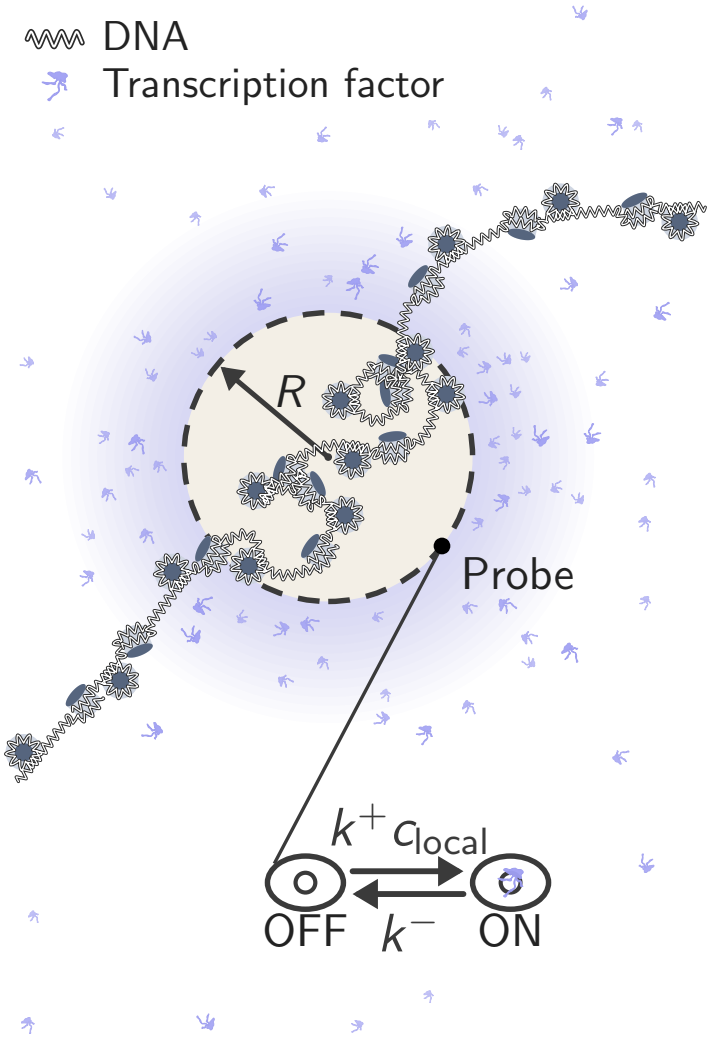
$$0 = \nabla^2 \left[f'(\phi) - \kappa \nabla^2 \phi \right]$$

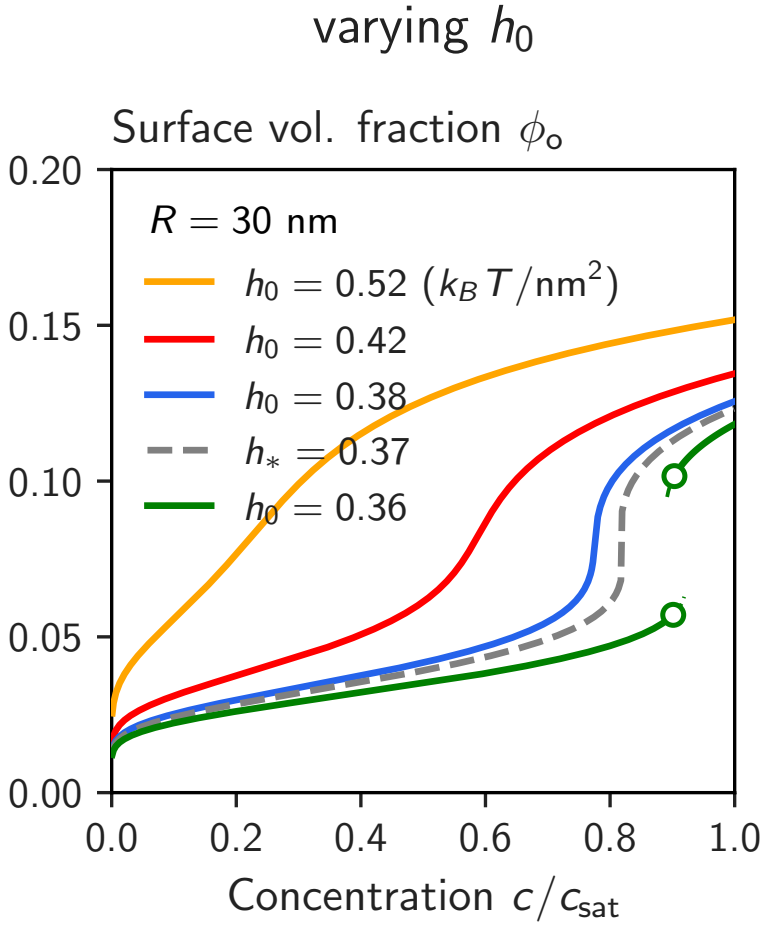
$$\kappa\, \partial_r \phi \,\big|_{r=R} = -h_0$$

ϕ	TF volume fraction
ν	TF / solvent volume ratio
χ	TF–TF attraction
μ	chemical potential
h_0	surface attraction
κ	interface stiffness

scene25$_1$

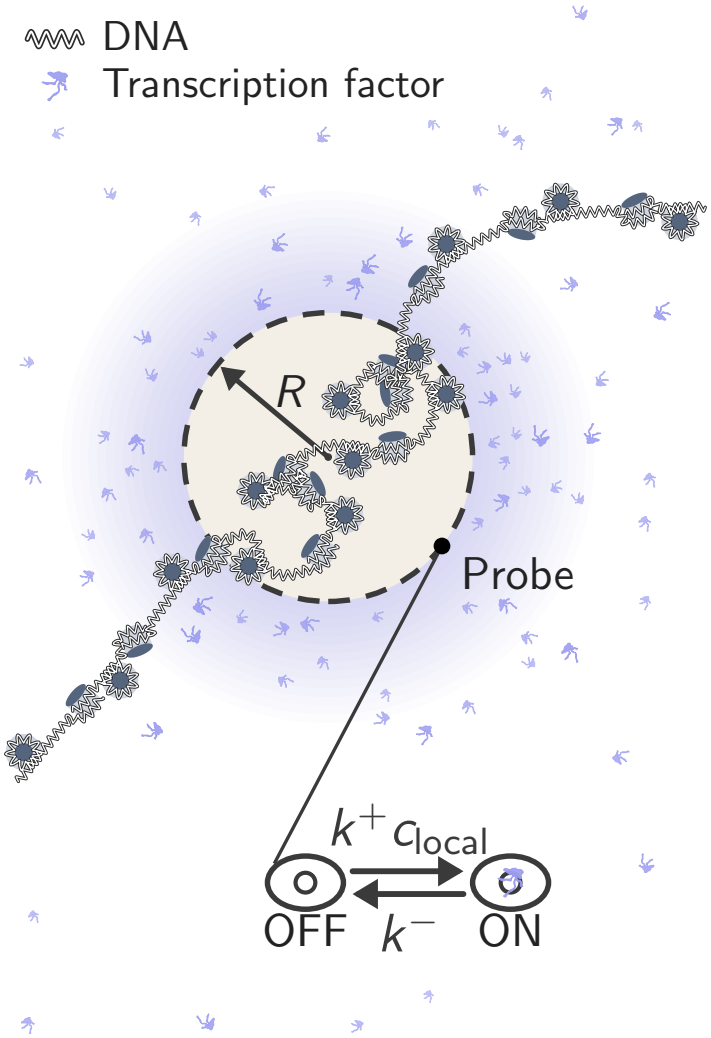
The prewetting transition: varying surface attraction

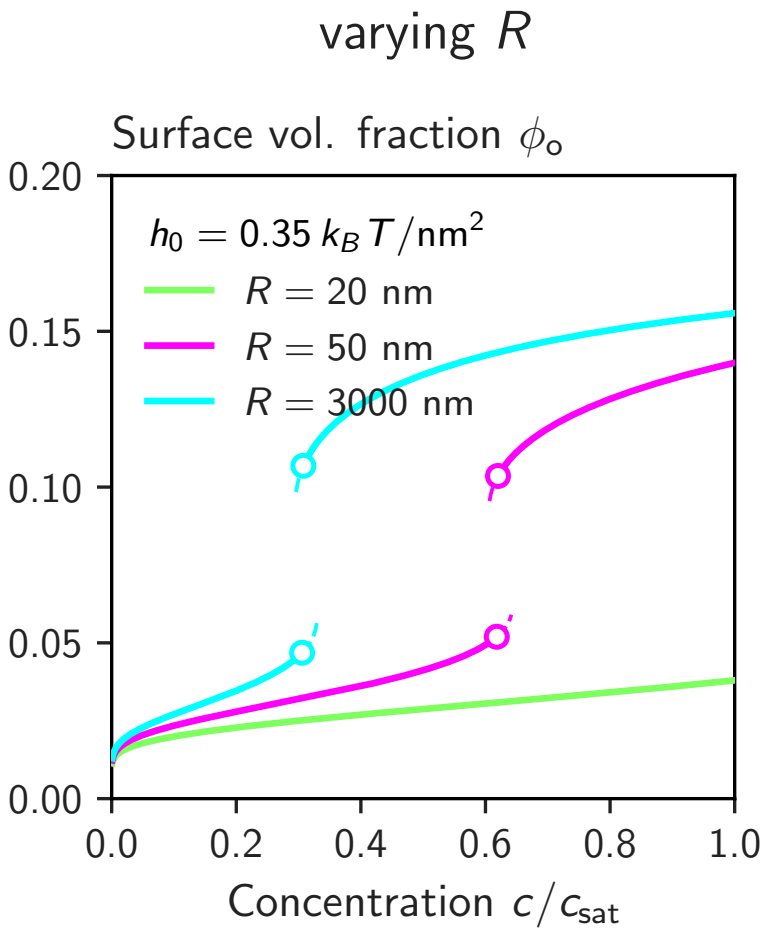




scene25_2

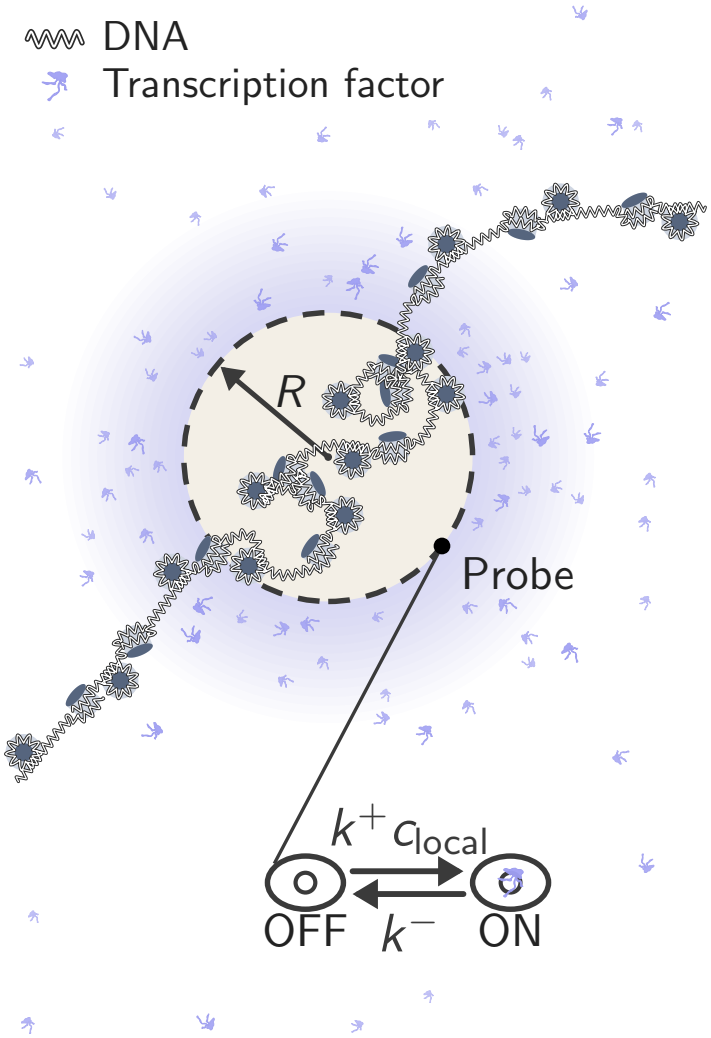
The prewetting transition: varying surface curvature





scene25$_3$

Quantifying sensitivity of local concentration



Free probe

$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_r(c)}$$

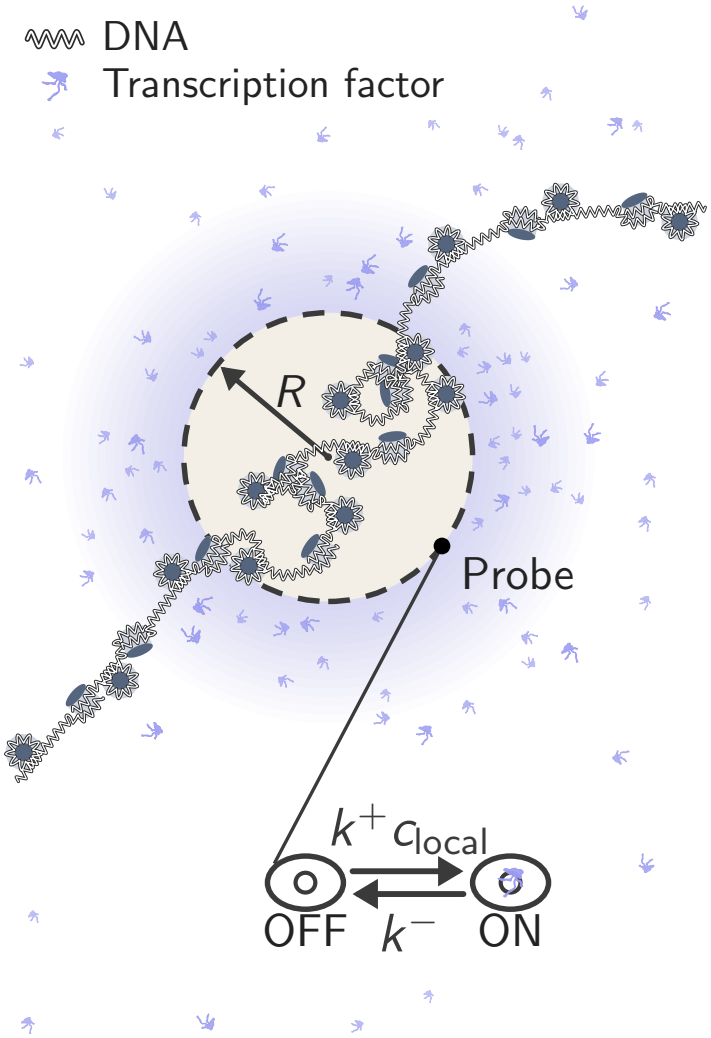
Probe on the anchor

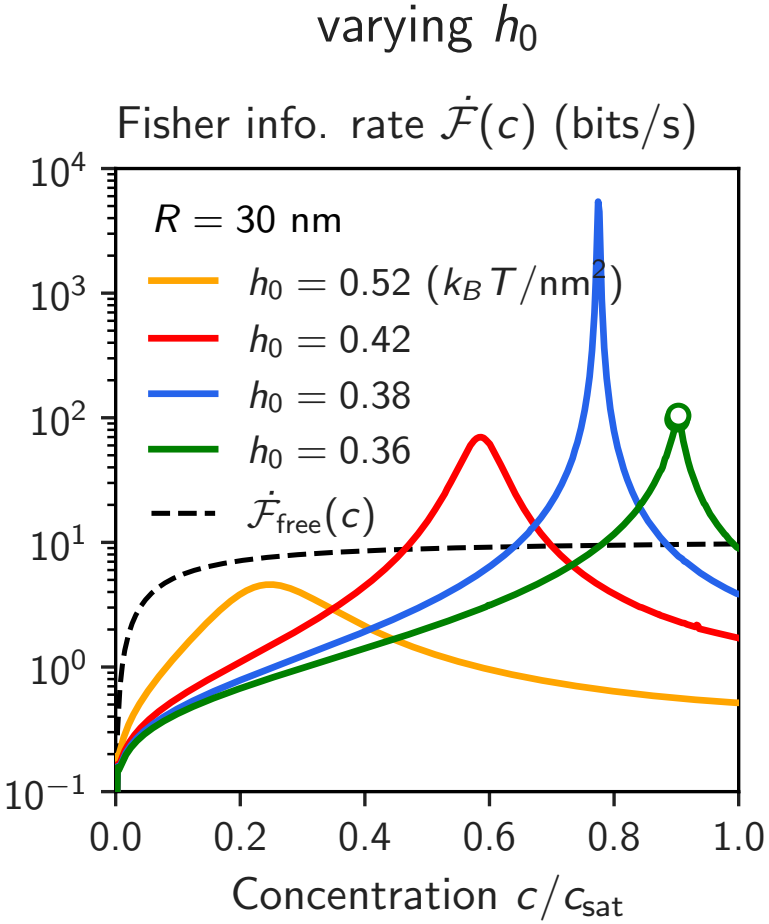
$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_r(\phi)}}_{\text{probe}} \underbrace{\left(\frac{d \log \phi}{d \log c}\right)^2}_{\text{surface}}$$

τ_r mean cycle time of the probe
ϕ TF volume fraction at the probe

scene26$_1$

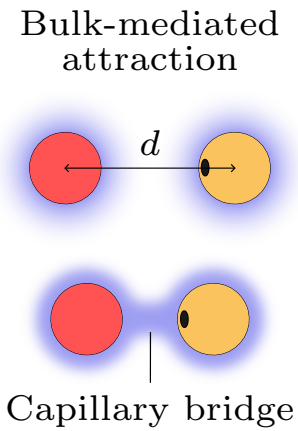
Quantifying sensitivity of local concentration

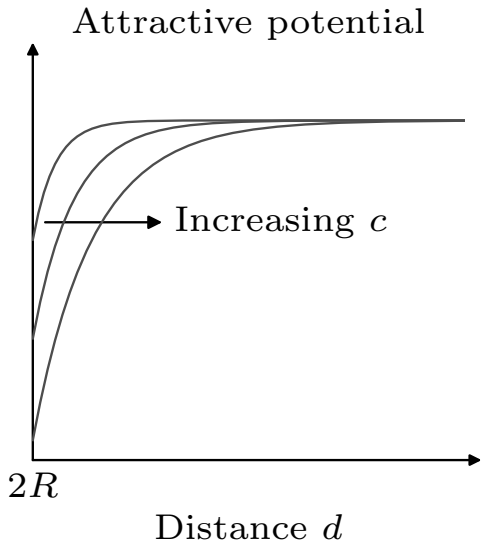


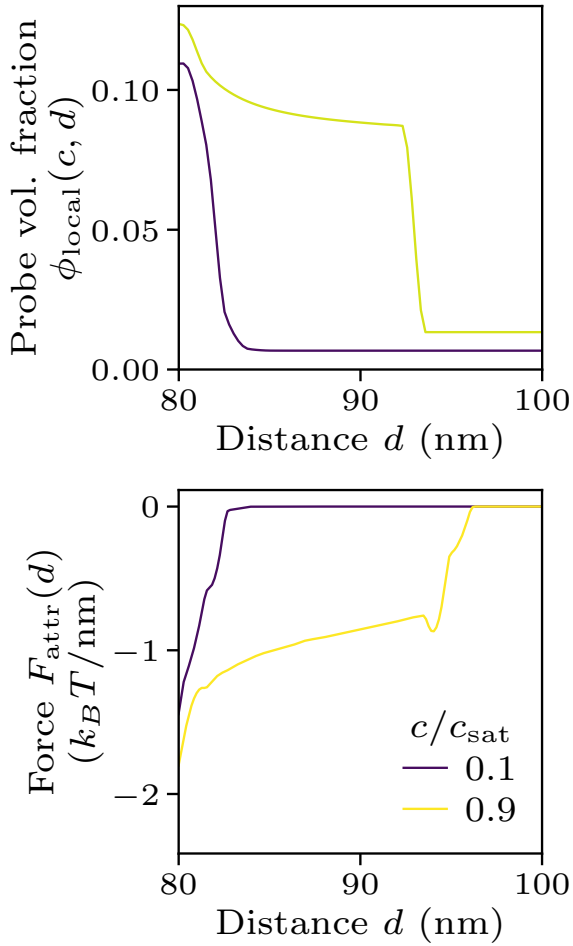


scene26_2

Bulk-mediated attraction

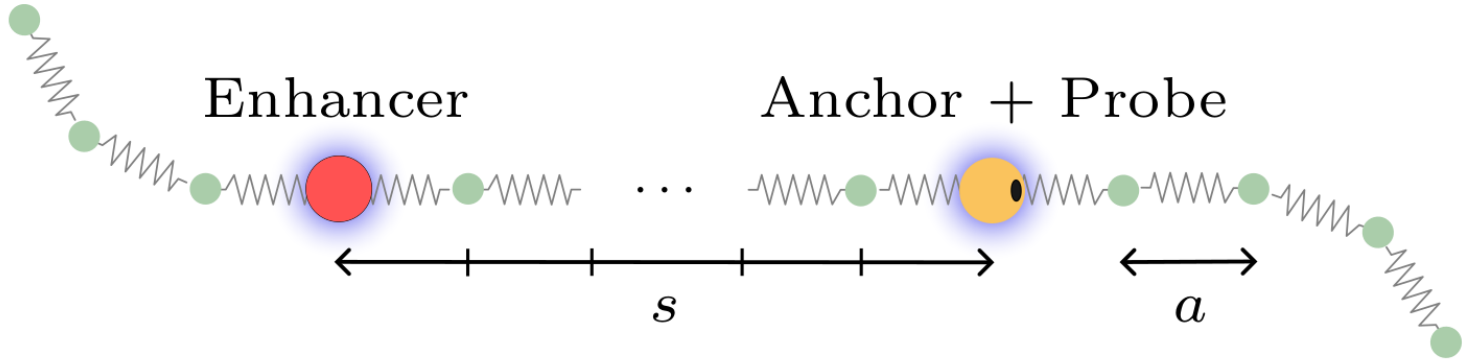






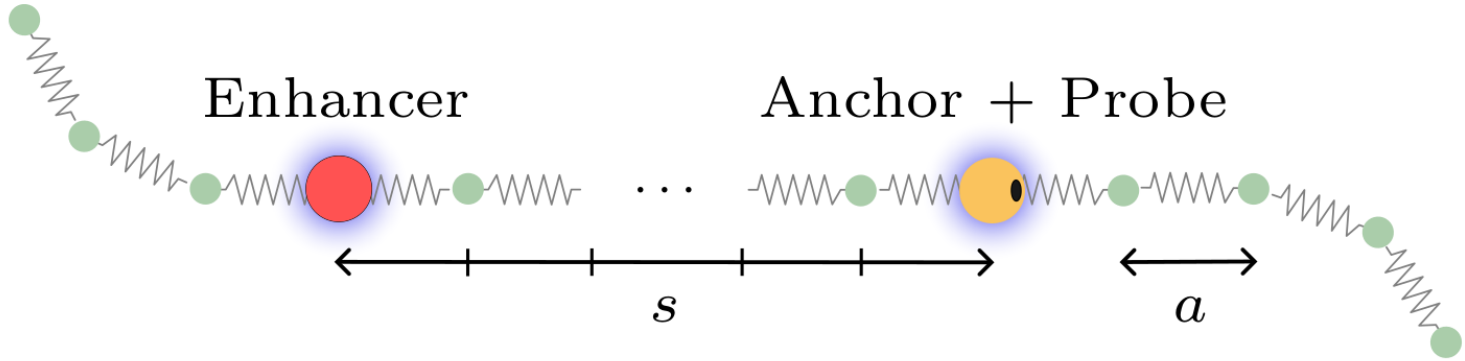
scene27

Modeling the polymer



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

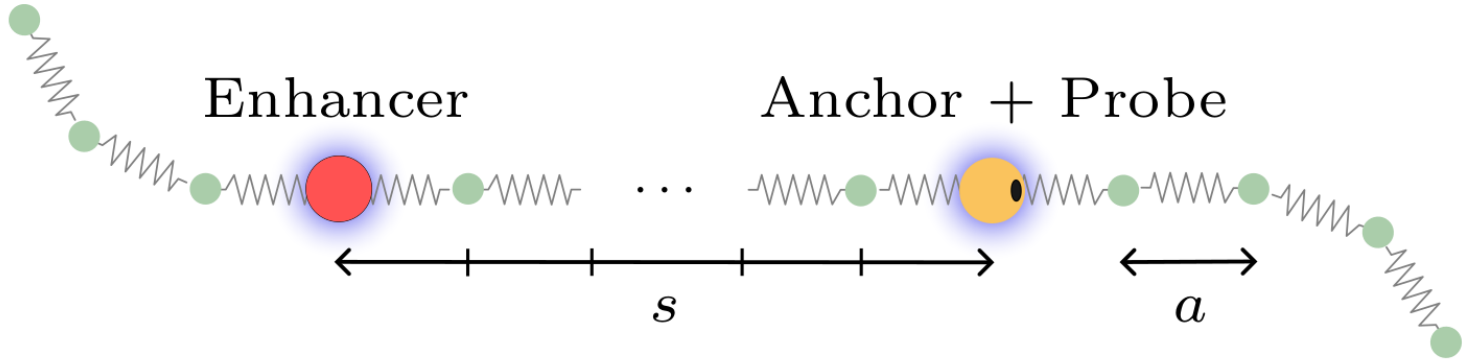
Modeling the polymer



$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

Modeling the polymer

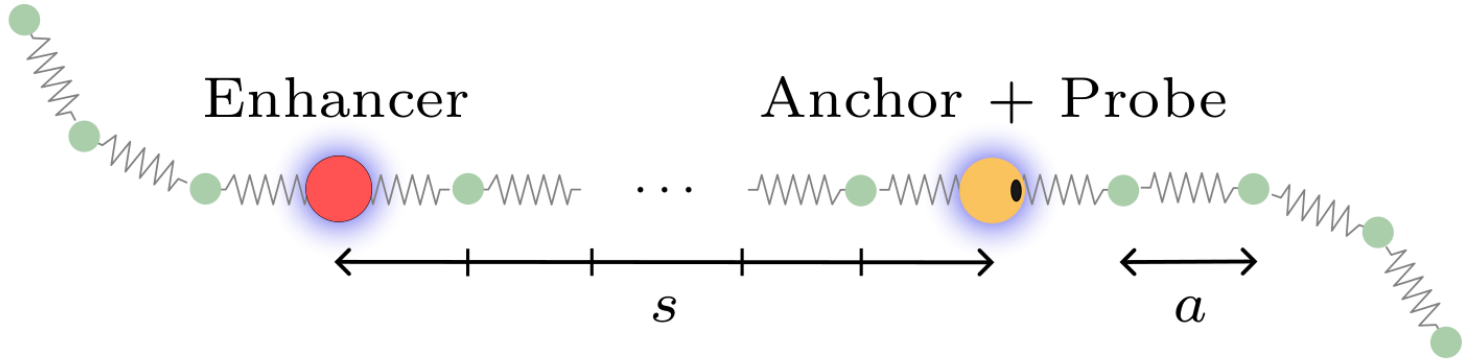


$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

Modeling the polymer



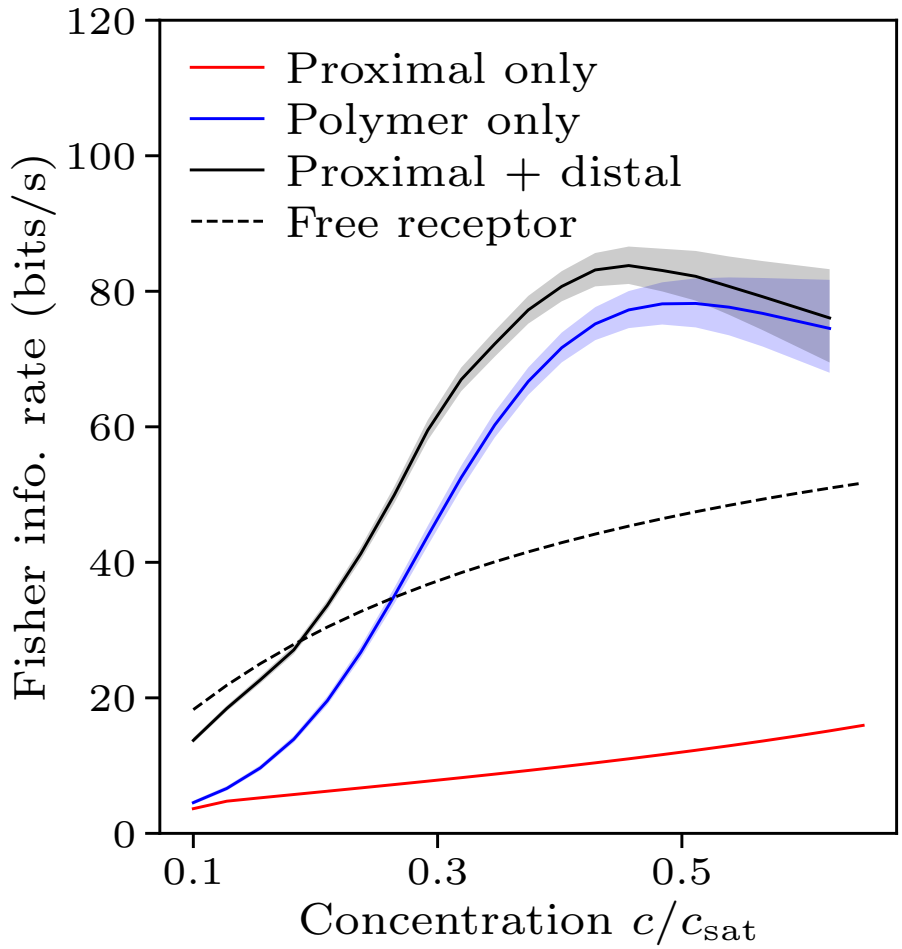
$$P_{\text{poly}}(d) \propto d^2 e^{-3d^2/2sa^2} \propto e^{-U_{\text{poly}}(d)/k_B T} \tag{1}$$

$$U_{\text{tot}}(d) = U_{\text{attr}}(d) + U_{\text{poly}}(d) \tag{2}$$

$$P_{\text{eq}}(d) \propto e^{-U_{\text{tot}}(d)/k_B T} \tag{3}$$

a: Kuhn bond length
s: genomic separation
d: euclidean distance

Sensitivity from polymer dynamics



scene30

Conclusions

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